CHAPTER 6

Wireless Networks and Mobile IP

We discussed wired networks in Chapter 5. In this chapter, we introduce wireless networks. We discuss several wireless technologies which include wireless LANs, and other wireless networks that span cellular telephone, satellites, and wireless access networks.

The nature of wireless LANs is different from that of wired LANs, as we will see in this chapter. The use of other wireless networks such as cellular telephony and satellite is increasing in the Internet. We discuss cellular phones and satellites as part of wireless technologies. We also devote one section in this chapter to mobile IP: the use of IP protocol with mobile hosts.

Wireless technology, in fact, covers both the data-link and the physical layer of the TCP/IP suite. We mostly concentrate on the issues related to the data-link layer and postpone issues related to the physical layer until Chapter 7.

This chapter is divided into three sections:

- The first section introduces wireless LANs and compares them with wired LANs, which we discussed in Chapter 5. We then discuss IEEE project 802.11, which is the dominant standard in wireless LANs. Next, we cover the Bluetooth LANs that are used as stand-alone LANs with many applications. We finally discuss WiMAX technology, which is the counterpart of last-mile wired networks such as DSL or cable.
- The second section discusses other wireless networks that can be categorized as wireless WANs or wireless broadband networks. We first discuss the channelization access method that is used in cellular telephones. We then discuss cellular telephones. We also deal with satellites that are integrated with the Internet connection.
- The third section covers mobile IP, which provides mobile access to the Internet. We first discuss addressing, a big issue in mobile networking. Then we discuss the three phases of mobile access. Finally, we mention the inefficiency in IP mobile.

6.1 WIRELESS LANs

Wireless communication is one of the fastest-growing technologies. The demand for connecting devices without the use of cables is increasing everywhere. Wireless LANs can be found on college campuses, in office buildings, and in many public areas.

6.1.1 Introduction

Before we discuss a specific protocol related to wireless LANs, let us talk about them in general.

Architectural Comparison

Let us first compare the architecture of wired and wireless LANs to give some idea of what we need to look for when we study wireless LANs.

Medium

The first difference we can see between a wired and a wireless LAN is the medium. In a wired LAN, we use wires to connect hosts. In Chapter 5, we saw that we moved from multiple access to point-to-point access through the generation of the Ethernet. In a switched LAN, with a link-layer switch, the communication between the hosts is point-to-point and full-duplex (bidirectional). In a wireless LAN, the medium is air, the signal is generally broadcast. When hosts in a wireless LAN communicate with each other, they are sharing the same medium (multiple access). In a very rare situation, we may be able to create a point-to-point communication between two wireless hosts by using a very limited bandwidth and two-directional antennas. Our discussion in this chapter, however, is about the multiple-access medium, which means we need to use MAC protocols.

Hosts

In a wired LAN, a host is always connected to its network at a point with a fixed link-layer address related to its network interface card (NIC). Of course, a host can move from one point in the Internet to another point. In this case, its link-layer address remains the same, but its network-layer address will change, as we see later in the chapter (Mobile IP section). However, before the host can use the services of the Internet, it needs to be physically connected to the Internet. In a wireless LAN, a host is not physically connected to the network; it can move freely (as we'll see) and can use the services provided by the network. Therefore, mobility in a wired network and wireless network are totally different issues, which we try to clarify in this chapter.

Isolated LANs

The concept of a wired isolated LAN also differs from that of a wireless isolated LAN. A wired isolated LAN is a set of hosts connected via a link-layer switch (in the recent generation of Ethernet). A wireless isolated LAN, called an **ad hoc network** in wireless LAN terminology, is a set of hosts that communicate freely with each other. The concept of a link-layer switch does not exist in wireless LANs. Figure 6.1 shows two isolated LANs, one wired and one wireless.

Connection to Other Networks

A wired LAN can be connected to another network or an internetwork such as the Internet using a router. A wireless LAN may be connected to a wired infrastructure network, to a wireless infrastructure network, or to another wireless LAN. The first situation is the one that we discuss in this section: connection of a wireless LAN to a wired infrastructure network. Figure 6.2 shows the two environments.

Figure 6.1 *Isolated LANs: wired versus wireless*

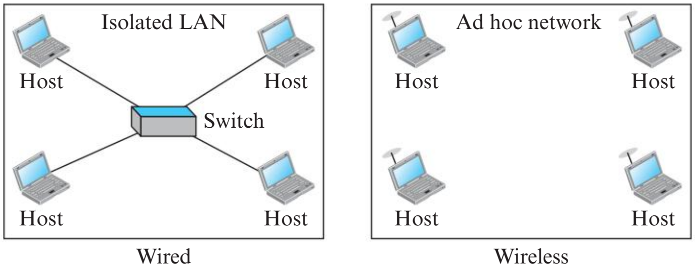
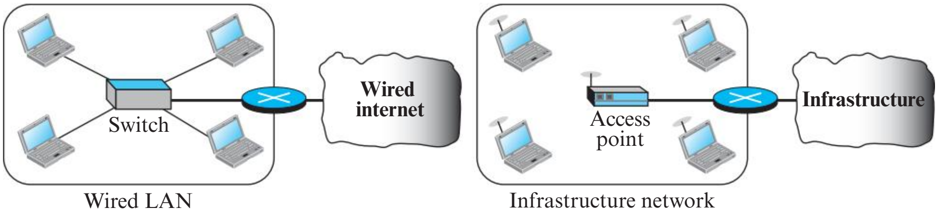


Figure 6.2 *Connection of a wired LAN and a wireless LAN to other networks*



In this case, the wireless LAN is referred to as an *infrastructure network,* and the connection to the wired infrastructure, such as the Internet, is done via a device called an **access point (AP)**. Note that the role of the access point is completely different from the role of a link-layer switch in the wired environment. An access point is gluing two different environments together: one wired and one wireless. Communication between the AP and the wireless host occurs in a wireless environment; communication between the AP and the infrastructure occurs in a wired environment.

Moving between Environments

The discussion above confirms what we learned in the previous chapter: a wired LAN or a wireless LAN operates only in the lower two layers of the TCP/IP protocol suite. This means that if we have a wired LAN in a building that is connected via a router or a modem to the Internet, all we need in order to move from the wired environment to a wireless environment is to change the network interface cards designed for wired environments to the ones designed for wireless environments and replace the link-layer switch with an access point. In this change, the link-layer addresses will change (because of changing NICs), but the network-layer addresses (IP addresses) will remain the same; we are moving from wired links to wireless links.

Characteristics

There are several characteristics of wireless LANs that either do not apply to wired LANs or the existence of which is negligible and can be ignored. We discuss some of these characteristics here to pave the way for discussing wireless LAN protocols.

Attenuation

The strength of electromagnetic signals decreases rapidly because the signal disperses in all directions; only a small portion of it reaches the receiver. The situation becomes worse with mobile senders that operate on batteries and normally have small power supplies.

Interference

Another issue is that a receiver may receive signals not only from the intended sender, but also from other senders if they are using the same frequency band.

Multipath Propagation

A receiver may receive more than one signal from the same sender because electromagnetic waves can be reflected back from obstacles such as walls, the ground, or objects. The result is that the receiver receives some different signals at different phases (because they travel different paths). This makes the signal less recognizable.

Error

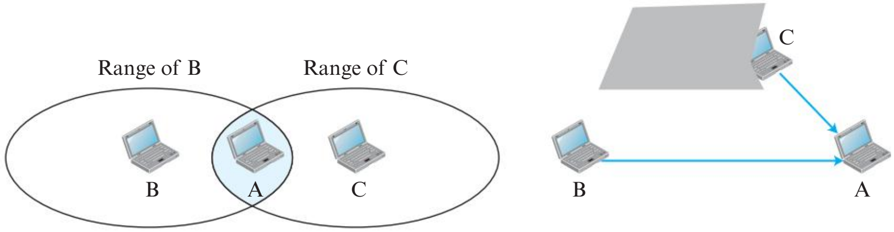
With the above characteristics of a wireless network, we can expect that errors and error detection are more serious issues in a wireless network than in a wired network. If we think about the error level as the measurement of **signal to noise ratio (SNR),** we can better understand why error detection and error correction and retransmission are more important in a wireless network. We discuss SNR in more detail in Chapter 7, but it is enough to say that it measures the ratio of good stuff to bad stuff (signal to noise). If SNR is high, it means that the signal is stronger than the noise (unwanted signal), so we may be able to convert the signal to actual data. On the other hand, when SNR is low, it means that the signal is corrupted by the noise and the data cannot be recovered.

Access Control

Maybe the most important issue we need to discuss in a wireless LAN is access control—how a wireless host can get access to the shared medium (air). We discussed in Chapter 5 that the Standard Ethernet used the CSMA/CD algorithm. In this method, each host contends to access the medium and sends its frame if it finds the medium idle. If a collision occurs, it is detected and the frame is sent again. Collision detection in CSMA/CD serves two purposes. If a collision is detected, it means that the frame has not been received and needs to be resent. If a collision is not detected, it is a kind of acknowledgment that the frame was received. The CSMA/CD algorithm does not work in wireless LAN for three reasons:

1. To detect a collision, a host needs to send and receive at the same time (sending the frame and receiving the collision signal), which means the host needs to work in a duplex mode. Wireless hosts do not have enough power to do so (the power is supplied by batteries). They can only send or receive at one time.
2. Because of the hidden station problem, in which a station may not be aware of another station's transmission due to some obstacles or range problems, collision may occur but not be detected. Figure 6.3 shows an example of the hidden station problem. Station B has a transmission range shown by the left oval (sphere in space); every station in this range can hear any signal transmitted by station B. Station C has a transmission range shown by the right oval (sphere in space); every station located in this range can hear any signal transmitted by C. Station C is

Figure 6.3 *Hidden station problem*



a. Stations B and C are not in each other's range.

b. Stations B and C are hidden from each other.

outside the transmission range of B; likewise, station B is outside the transmission range of C. Station A, however, is in the area covered by both B and C; it can hear any signal transmitted by B or C. The figure also shows that the hidden station problem may also occur due to an obstacle.

Assume that station B is sending data to station A. In the middle of this transmission, station C also has data to send to station A. However, station C is out of B's range and transmissions from B cannot reach C. Therefore C thinks the medium is free. Station C sends its data to A, which results in a collision at A because this station is receiving data from both B and C. In this case, we say that stations B and C are hidden from each other with respect to A. Hidden stations can reduce the capacity of the network because of the possibility of collision.

3. The distance between stations can be great. Signal fading could prevent a station at one end from hearing a collision at the other end.

To overcome the above three problems, Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) was invented for wireless LAN. We discuss this later.

6.1.2 IEEE 802.11 Project

IEEE has defined the specifications for a wireless LAN, called IEEE 802.11, which covers the physical and data-link layers. In some countries, including the United States, the public uses the term *WiFi* (short for wireless fidelity) as a synonym for *wireless LAN*. WiFi, however, is a wireless LAN that is certified by the WiFi Alliance, a global, nonprofit industry association of more than 300 member companies devoted to promoting the growth of wireless LANs.

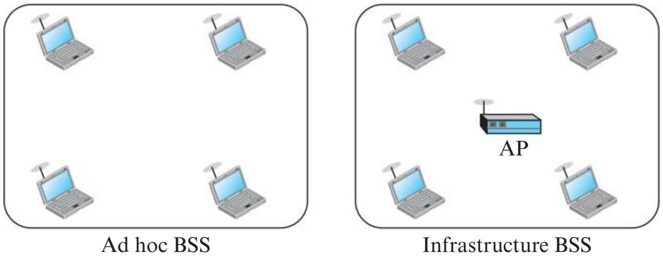
Architecture

The standard defines two kinds of services: the basic service set (BSS) and the extended service set (ESS).

Basic Service Set

IEEE 802.11 defines the **basic service set (BSS)** as the building blocks of a wireless LAN. A basic service set is made of stationary or mobile wireless stations and an optional central base station, known as the access point (AP). Figure 6.4 shows two sets in this standard.

Figure 6.4 *Basic service sets (BSSs)*

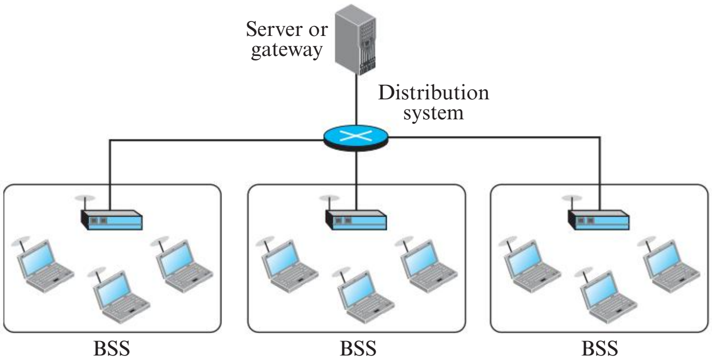


The BSS without an AP is a stand-alone network and cannot send data to other BSSs. It is called an *ad hoc architecture*. In this architecture, stations can form a network without the need of an AP; they can locate one another and agree to be part of a BSS. A BSS with an AP is sometimes referred to as an *infrastructure* BSS.

Extended Service Set

An **extended service set (ESS)** is made up of two or more BSSs with APs. In this case, the BSSs are connected through a *distribution system,* which is a wired or a wireless network. The distribution system connects the APs in the BSSs. IEEE 802.11 does not restrict the distribution system; it can be any IEEE LAN such as an Ethernet. Note that the extended service set uses two types of stations: mobile and stationary. The mobile stations are normal stations inside a BSS. The stationary stations are AP stations that are part of a wired LAN. Figure 6.5 shows an ESS.

Figure 6.5 *Extended service set (ESS)*



When BSSs are connected, the stations within reach of one another can communicate without the use of an AP. However, communication between a station in a BSS and the outside BSS occurs via the AP. The idea is similar to communication in a cellular network (discussed later) if we consider each BSS to be a cell and each AP to be a base station. Note that a mobile station can belong to more than one BSS at the same time.

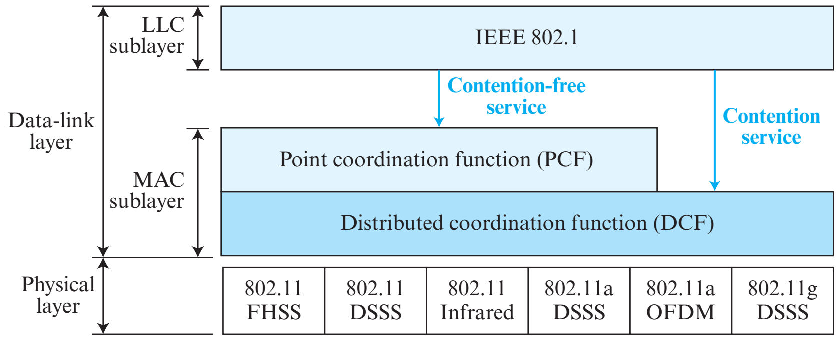
Station Types

IEEE 802.11 defines three types of stations based on their mobility in a wireless LAN: *no-transition*, *BSS-transition*, and *ESS-transition* mobility. A station with no-transition mobility is either stationary (not moving) or moving only inside a BSS. A station with BSS-transition mobility can move from one BSS to another, but the movement is confined inside one ESS. A station with ESS-transition mobility can move from one ESS to another. However, IEEE 802.11 does not guarantee that communication is continuous during the move.

MAC Sublayer

IEEE 802.11 defines two MAC sublayers: the distributed coordination function (DCF) and point coordination function (PCF). Figure 6.6 shows the relationship between the two MAC sublayers, the LLC sublayer, and the physical layer. We discuss the physical layer implementations later in the chapter and will now concentrate on the MAC sublayer.

Figure 6.6 *MAC layers in IEEE 802.11 standard*



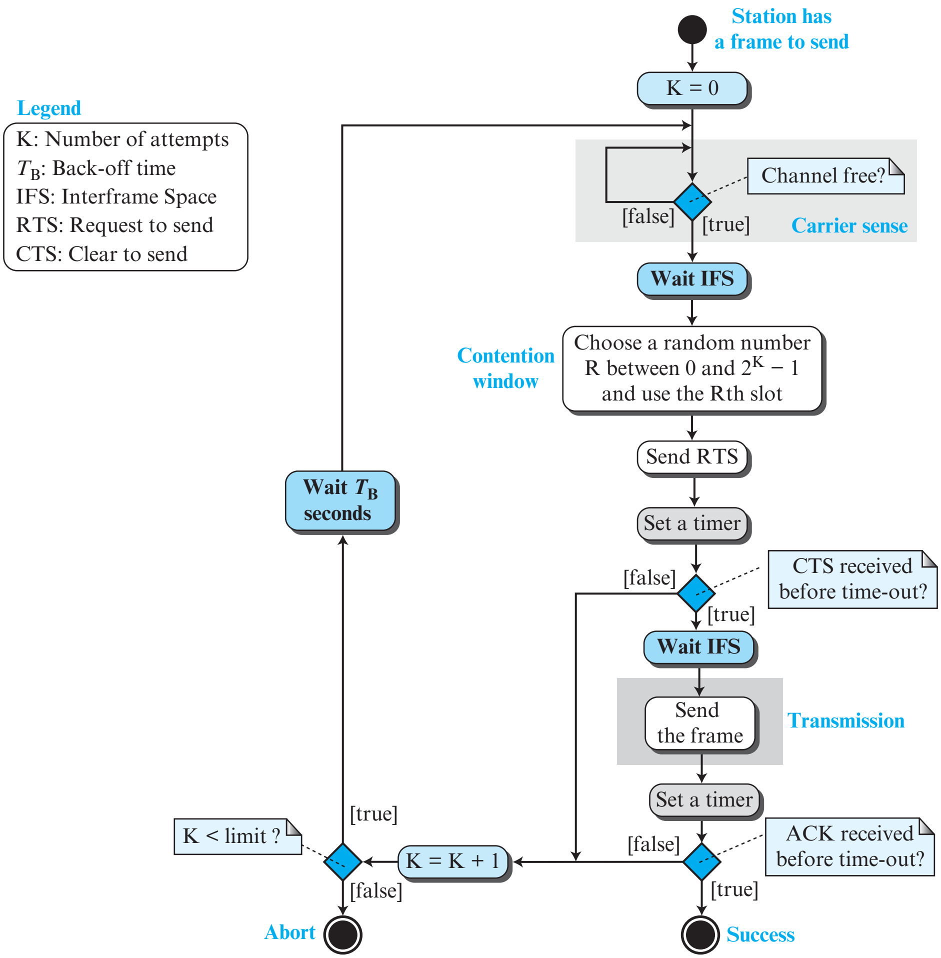
Distributed Coordination Function

One of the two protocols defined by IEEE at the MAC sublayer is called the **distributed coordination function (DCF)**. DCF uses CSMA/CA as the access method.

CSMA/CA Since we need to avoid collisions on wireless networks because they cannot be detected, carrier sense multiple access with collision avoidance (CSMA/CA) was invented for this network. Collisions are avoided through the use of CSMA/CA's three strategies: the interframe space, the contention window, and acknowledgments, as shown in Figure 6.7. We discuss RTS and CTS frames later.

- ***Interframe Space (IFS).*** First, collisions are avoided by deferring transmission even if the channel is found idle. When an idle channel is found, the station does not send immediately. It waits for a period of time called the *interframe space* or *IFS*. Even though the channel may appear idle when it is sensed, a distant station may have already started transmitting. The distant station's signal has not yet reached this station. The IFS time allows the front of the transmitted signal by the distant station to reach this station. After waiting an IFS time, if the channel is still idle, the station can

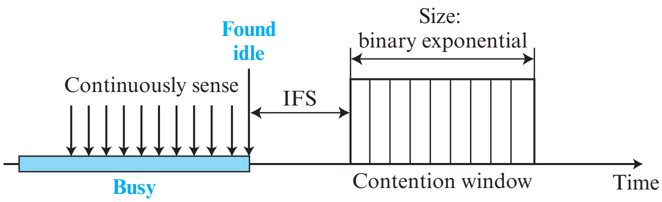
Figure 6.7 *Flow diagram of CSMA/CA*



send, but it still needs to wait a time equal to the contention time (described next). The IFS variable can also be used to prioritize stations or frame types. For example, a station that is assigned a shorter IFS has a higher priority.

- ***Contention Window.*** The contention window is an amount of time divided into slots. A station that is ready to send chooses a random number of slots as its wait time. The number of slots in the window changes according to the binary exponential back-off strategy. This means that it is set to one slot the first time and then doubles each time the station cannot detect an idle channel after the IFS time. This is very similar to the *p*-persistent method except that a random outcome defines the number of slots taken by the waiting station. One interesting point about the contention window is that the station needs to sense the channel after each time slot. However, if the station finds the channel busy, it does not restart the process; it just stops the timer and restarts it when the channel is sensed as idle. This gives priority to the station with the longest waiting time. See Figure 6.8

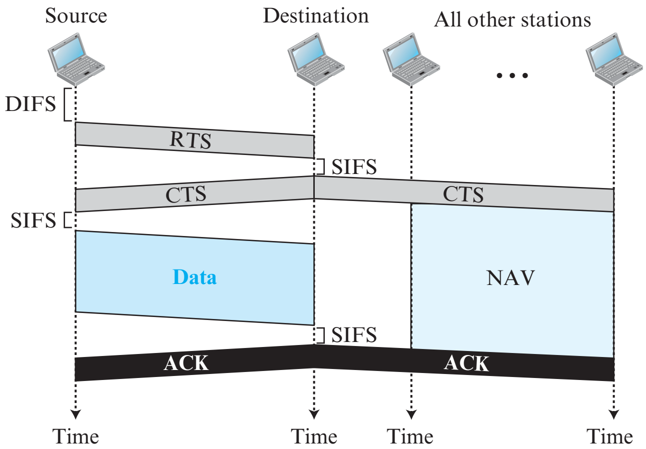
Figure 6.8 *Contention window*



- ***Acknowledgment.*** With all these precautions, there still may be a collision resulting in destroyed data. In addition, the data may be corrupted during the transmission. The positive acknowledgment and the time-out timer can help guarantee that the receiver has received the frame.

Frame Exchange Time Line Figure 6.9 shows the exchange of data and control frames in time.

Figure 6.9 *CSMA/CA and NAV*



1. Before sending a frame, the source station senses the medium by checking the energy level at the carrier frequency.
   a. The channel uses a persistence strategy with back-off until the channel is idle.
   b. After the station is found to be idle, the station waits for a period of time called the **DCF interframe space (DIFS);** then the station sends a control frame called the *request to send (RTS).*
2. After receiving the RTS and waiting a period of time called the **short interframe space (SIFS),** the destination station sends a control frame, called the *clear to*

send (CTS), to the source station. This control frame indicates that the destination station is ready to receive data.

3. The source station sends data after waiting an amount of time equal to SIFS.
4. The destination station, after waiting an amount of time equal to SIFS, sends an acknowledgment to show that the frame has been received. Acknowledgment is needed in this protocol because the station does not have any means to check for the successful arrival of its data at the destination. On the other hand, the lack of collision in CSMA/CD is a kind of indication to the source that data have arrived.

Network Allocation Vector How do other stations defer sending their data if one station acquires access? In other words, how is the *collision avoidance* aspect of this protocol accomplished? The key is a feature called NAV.

When a station sends an RTS frame, it includes the duration of time that it needs to occupy the channel. The stations that are affected by this transmission create a timer called a **network allocation vector (NAV)** that shows how much time must pass before these stations are allowed to check the channel for idleness. Each time a station accesses the system and sends an RTS frame, other stations start their NAV. In other words, each station, before sensing the physical medium to see if it is idle, first checks its NAV to see if it has expired. Figure 6.9 shows the idea of NAV.

Collision During Handshaking What happens if there is a collision during the time when RTS or CTS control frames are in transition, often called the handshaking period? Two or more stations may try to send RTS frames at the same time. These control frames may collide. However, because there is no mechanism for collision detection, the sender assumes there has been a collision if it has not received a CTS frame from the receiver. The back-off strategy is employed, and the sender tries again.

Hidden-Station Problem The solution to the hidden station problem is the use of the handshake frames (RTS and CTS). Figure 6.3 also shows that the RTS message from B reaches A, but not C. However, because both B and C are within the range of A, the CTS message, which contains the duration of data transmission from B to A, reaches C. Station C knows that some hidden station is using the channel and refrains from transmitting until that duration is over.

Point Coordination Function (PCF)

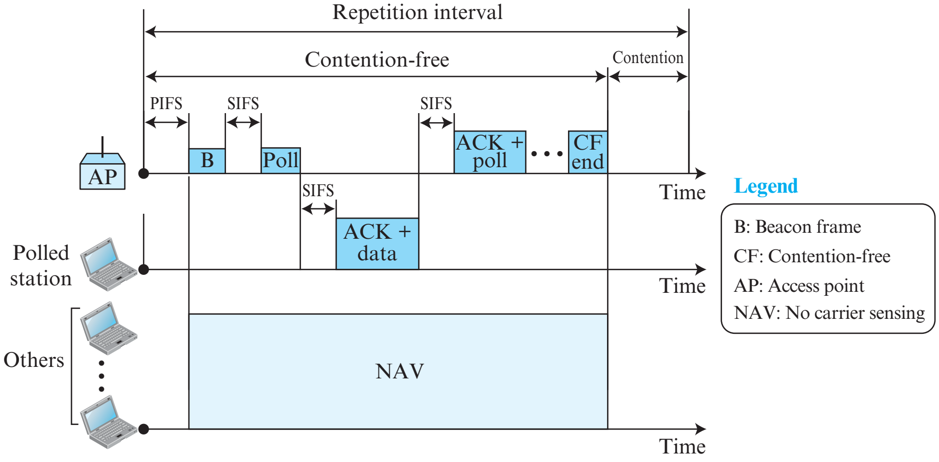
The **point coordination function (PCF)** is an optional access method that can be implemented in an infrastructure network (not in an ad hoc network). It is implemented on top of the DCF and is used mostly for time-sensitive transmission.

PCF has a centralized, contention-free polling access method, which we discussed in Chapter 5. The AP performs polling for stations that are capable of being polled. The stations are polled one after another, sending any data they have to the AP.

To give priority to PCF over DCF, another interframe space, PIFS, has been defined. PIFS (PCF IFS) is shorter than the DIFS. This means that if, at the same time, a station wants to use only DCF and an AP wants to use PCF, the AP has priority.

Due to the priority of PCF over DCF, stations that only use DCF may not gain access to the medium. To prevent this, a repetition interval has been designed to cover both contention-free PCF and contention-based DCF traffic. The *repetition interval,* which is repeated continuously, starts with a special control frame, called a **beacon frame.** When the stations hear the beacon frame, they start their NAV for the duration of the contention-free period of the repetition interval. Figure 6.10 shows an example of a repetition interval.

Figure 6.10 *Example of repetition interval*



During the repetition interval, the PC (point controller) can send a poll frame, receive data, send an ACK, receive an ACK, or do any combination of these (802.11 uses piggybacking). At the end of the contention-free period, the PC sends a CF end (contention-free end) frame to allow the contention-based stations to use the medium.

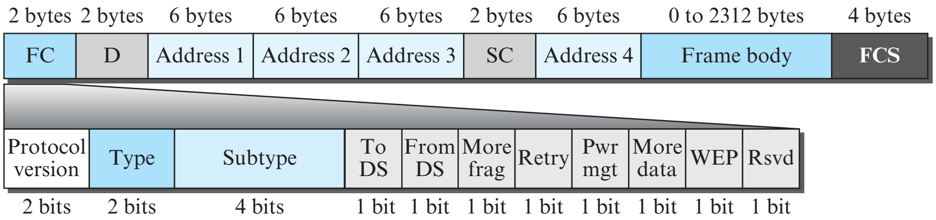
Fragmentation

The wireless environment is very noisy, so frames are often corrupted. A corrupt frame has to be retransmitted. The protocol, therefore, recommends fragmentation—the division of a large frame into smaller ones. It is more efficient to resend a small frame than a large one.

Frame Format

The MAC layer frame consists of nine fields, as shown in Figure 6.11.

Figure 6.11 *Frame format*



- ***Frame control (FC).*** The FC field is 2 bytes long and defines the type of frame and some control information. Table 6.1 describes the subfields. We will discuss each frame type later in this chapter.

Table 6.1 *Subfields in FC field*

Field	*Explanation*
Version	Current version is 0
Type	Type of information: management (00), control (01), or data (10)
Subtype	Subtype of each type (see Table 6.2)
To DS	Defined later
From DS	Defined later
More flag	When set to 1, means more fragments
Retry	When set to 1, means retransmitted frame
Pwr mgt	When set to 1, means station is in power management mode
More data	When set to 1, means station has more data to send
WEP	Wired equivalent privacy (encryption implemented)
Rsvd	Reserved

- ***D.*** This field defines the duration of the transmission that is used to set the value of NAV. In one control frame, it defines the ID of the frame.
- ***Addresses.*** There are four address fields, each 6 bytes long. The meaning of each address field depends on the value of the *To DS* and *From DS* subfields and will be discussed later.
- ***Sequence control.*** This field, often called the SC field, defines a 16-bit value. The first four bits define the fragment number; the last 12 bits define the sequence number, which is the same in all fragments.
- ***Frame body.*** This field, which can be between 0 and 2312 bytes, contains information based on the type and the subtype defined in the FC field.
- ***FCS.*** The FCS field is 4 bytes long and contains a CRC-32 error-detection sequence.

Frame Types

A wireless LAN defined by IEEE 802.11 has three categories of frames: management frames, control frames, and data frames.

Management Frames Management frames are used for the initial communication between stations and access points.

Control Frames Control frames are used for accessing the channel and acknowledging frames. Figure 6.12 shows the format.

Figure 6.12 *Control frames*

For control frames the value of the type field is 01; the values of the subtype fields for frames we have discussed are shown in Table 6.2.

Table 6.2 *Values of subfields in control frames*

Subtype	*Meaning*
1011	Request to send (RTS)
1100	Clear to send (CTS)
1101	Acknowledgment (ACK)

Data Frames Data frames are used for carrying data and control information.

Addressing Mechanism

The IEEE 802.11 addressing mechanism specifies four cases, defined by the value of the two flags in the FC field, *To DS* and *From DS*. Each flag can be either 0 or 1, resulting in four different situations. The interpretation of the four addresses (address 1 to address 4) in the MAC frame depends on the value of these flags, as shown in Table 6.3.

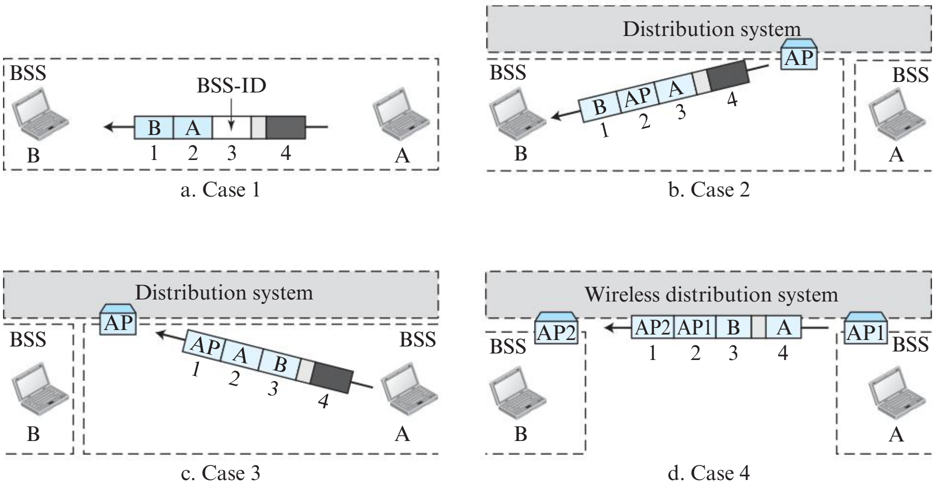
Table 6.3 *Addresses*

To DS	*From DS*	*Address 1*	*Address 2*	*Address 3*	*Address 4*
0	0	Destination	Source	BSS ID	N/A
0	1	Destination	Sending AP	Source	N/A
1	0	Receiving AP	Source	Destination	N/A
1	1	Receiving AP	Sending AP	Destination	Source

Note that address 1 is always the address of the next device that the frame will visit. Address 2 is always the address of the previous device that the frame has left. Address 3 is the address of the final destination station if it is not defined by address 1 or the original source station if is not defined by address 2. Address 4 is the original source when the distribution system is also wireless.

- **Case 1:00** In this case, *To DS* = 0 and *From DS* = 0. This means that the frame is not going to a distribution system (*To DS* = 0) and is not coming from a distribution system (*From DS* = 0). The frame is going from one station in a BSS to another without passing through the distribution system. The addresses are shown in Figure 6.13.
- **Case 2:01** In this case, *To DS* = 0 and *From DS* = 1. This means that the frame is coming from a distribution system (*From DS* = 1). The frame is coming from an AP and going to a station. The addresses are as shown in Figure 6.13. Note that address 3 contains the original sender of the frame (in another BSS).
- **Case 3:10** In this case, *To DS* = 1 and *From DS* = 0. This means that the frame is going to a distribution system (*To DS* = 1). The frame is going from a station to an AP. The ACK is sent to the original station. The addresses are as shown in Figure 6.13. Note that address 3 contains the final destination of the frame in the distribution system.
- **Case 4:11** In this case, *To DS* = 1 and *From DS* = 1. This is the case in which the distribution system is also wireless. The frame is going from one AP to another AP

Figure 6.13 *Addressing mechanisms*

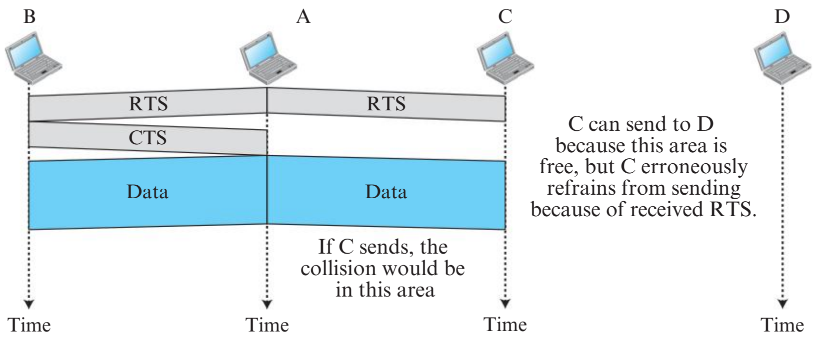


in a wireless distribution system. Here, we need four addresses to define the original sender, the final destination, and two intermediate APs. Figure 6.13 shows the situation.

Exposed Station Problem

We discussed how to solve the hidden station problem. A similar problem is called the *exposed station problem*. In this problem a station refrains from using a channel when it is, in fact, available. In Figure 6.14, station A is transmitting to station B. Station C has some data to send to station D, which can be sent without interfering with the transmission from A to B. However, station C is exposed to transmission from A; it hears what A is sending and thus refrains from sending. In other words, C is too conservative and wastes the capacity of the channel. The handshaking messages RTS and CTS cannot help in this case. Station C hears the RTS from A and refrains from sending, even

Figure 6.14 *Exposed station problem*



though the communication between C and D cannot cause a collision in the zone between A and C; station C cannot know that station A's transmission does not affect the zone between C and D.

Physical Layer

We discuss six specifications, as shown in Table 6.4. All implementations, except the infrared, operate in the *industrial, scientific,* and *medical (ISM)* band, which defines three unlicensed bands in the three ranges 902–928 MHz, 2.400–4.835 GHz, and 5.725–5.850 GHz.

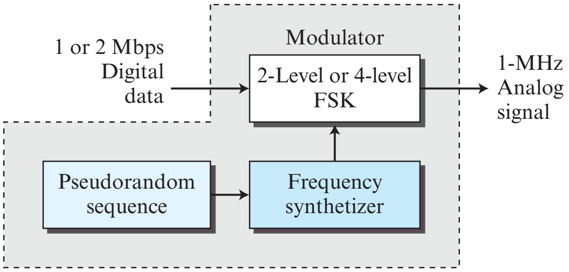
Table 6.4 *Specifications*

IEEE	*Technique*	*Band*	*Modulation*	*Rate (Mbps)*
802.11	FHSS	2.400–4.835 GHz	FSK	1 and 2
	DSSS	2.400–4.835 GHz	PSK	1 and 2
	None	Infrared	PPM	1 and 2
802.11a	OFDM	5.725–5.850 GHz	PSK or QAM	6 to 54
802.11b	DSSS	2.400–4.835 GHz	PSK	5.5 and 11
802.11g	OFDM	2.400–4.835 GHz	Different	22 and 54
802.11n	OFDM	5.725–5.850 GHz	Different	600

IEEE 802.11 FHSS

IEEE 802.11 FHSS uses the frequency-hopping spread spectrum (FHSS) method, as discussed in Chapter 7. FHSS uses the 2.400–4.835 GHz ISM band. The band is divided into 79 subbands of 1 MHz (and some guard bands). A pseudorandom number generator selects the hopping sequence. The modulation technique in this specification is either two-level FSK or four-level FSK with 1 or 2 bits/baud, which results in a data rate of 1 or 2 Mbps, as shown in Figure 6.15.

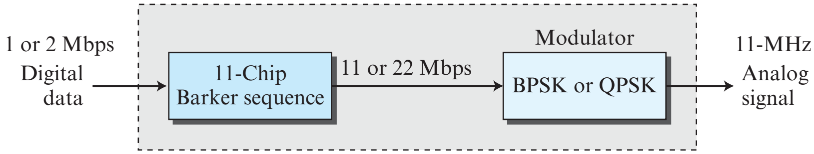
Figure 6.15 *Physical layer of IEEE 802.11 FHSS*



IEEE 802.11 DSSS

IEEE 802.11 **DSSS** uses the direct sequence spread spectrum (DSSS) method, as discussed in Chapter 7. DSSS uses the 2.400–4.835 GHz ISM band. The modulation technique in this specification is PSK at 1 Mbaud/s. The system allows 1 or 2 bits/baud (BPSK or QPSK), which results in a data rate of 1 or 2 Mbps, as shown in Figure 6.16.

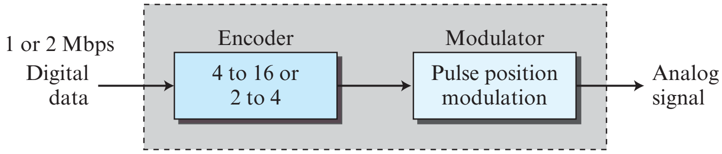
Figure 6.16 *Physical layer of IEEE 802.11 DSSS*



IEEE 802.11 Infrared

IEEE 802.11 infrared uses infrared light in the range of 800 to 950 nm. The modulation technique is called **pulse position modulation (PPM).** For a 1-Mbps data rate, a 4-bit sequence is first mapped into a 16-bit sequence in which only one bit is set to 1 and the rest are set to 0. For a 2-Mbps data rate, a 2-bit sequence is first mapped into a 4-bit sequence in which only one bit is set to 1 and the rest are set to 0. The mapped sequences are then converted to optical signals; the presence of light specifies 1, the absence of light specifies 0. See Figure 6.17.

Figure 6.17 *Physical layer of IEEE 802.11 infrared*



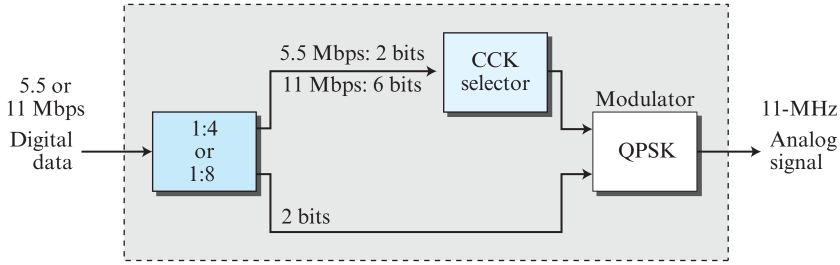
IEEE 802.11a OFDM

IEEE 802.11a OFDM describes the **orthogonal frequency-division multiplexing (OFDM)** method for signal generation in a 5.725–5.850 GHz ISM band. OFDM is similar to FDM, as discussed in Chapter 7, with one major difference: All the subbands are used by one source at a given time. Sources contend with one another at the data-link layer for access. The band is divided into 52 subbands, with 48 subbands for sending 48 groups of bits at a time and 4 subbands for control information. Dividing the band into subbands diminishes the effects of interference. If the subbands are used randomly, security can also be increased. OFDM uses PSK and QAM for modulation. The common data rates are 18 Mbps (PSK) and 54 Mbps (QAM).

IEEE 802.11b DSSS

IEEE 802.11b DSSS describes the **high-rate direct-sequence spread spectrum (HR-DSSS)** method for signal generation in the 2.400–4.835 GHz ISM band. HR-DSSS is similar to DSSS except for the encoding method, which is called **complementary code keying (CCK).** CCK encodes 4 or 8 bits to one CCK symbol. To be backward compatible with DSSS, HR-DSSS defines four data rates: 1, 2, 5.5, and 11 Mbps. The first two use the same modulation techniques as DSSS. The 5.5-Mbps version uses BPSK and transmits at 1.375 Mbaud/s with 4-bit CCK encoding. The 11-Mbps version uses QPSK and transmits at 1.375 Mbps with 8-bit CCK encoding. Figure 6.18 shows the modulation technique for this standard.

Figure 6.18 *Physical layer of IEEE 802.11b*



IEEE 802.11g

This new specification defines forward error correction and OFDM using the 2.400–4.835 GHz ISM band. The modulation technique achieves a 22- or 54-Mbps data rate. It is backward-compatible with 802.11b, but the modulation technique is OFDM.

IEEE 802.11n

An upgrade to the 802.11 project is called 802.11n (the next generation of wireless LAN). The goal is to increase the throughput of 802.11 wireless LANs. The new standard emphasizes not only the higher bit rate but also eliminating some unnecessary overhead. The standard uses what is called **MIMO (multiple-input multiple-output)** to overcome the noise problem in wireless LANs. The idea is that if we can send multiple output signals and receive multiple input signals, we are in the better position to eliminate noise. Some implementations of this project have reached up to 600 Mbps data rate.

6.1.3 Bluetooth

Bluetooth is a wireless LAN technology designed to connect devices of different functions such as telephones, notebooks, computers (desktop and laptop), cameras, printers, and even coffee makers when they are at a short distance from each other. A Bluetooth LAN is an ad hoc network, which means that the network is formed spontaneously; the devices, sometimes called gadgets, find each other and make a network called a piconet. A Bluetooth LAN can even be connected to the Internet if one of the gadgets has this capability. A Bluetooth LAN, by nature, cannot be large. If there are many gadgets that try to connect, there is chaos.

Bluetooth technology has several applications. Peripheral devices such as a wireless mouse or keyboard can communicate with the computer through this technology. Monitoring devices can communicate with sensor devices in a small health care center. Home security devices can use this technology to connect different sensors to the main security controller. Conference attendees can synchronize their laptop computers at a conference.

Bluetooth was originally started as a project by the Ericsson Company. It is named for Harald Blaatand, the king of Denmark (940-981) who united Denmark and Norway. *Blaatand* translates to *Bluetooth* in English.

Today, Bluetooth technology is the implementation of a protocol defined by the IEEE 802.15 standard. The standard defines a wireless personal-area network (PAN) operable in an area the size of a room or a hall.

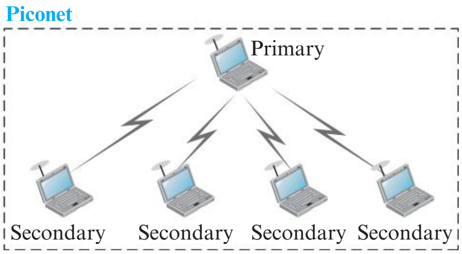
Architecture

Bluetooth defines two types of networks: piconet and scatternet.

Piconets

A Bluetooth network is called a **piconet,** or a small net. A piconet can have up to eight stations, one of which is called the *primary*; the rest are called *secondaries.* All the secondary stations synchronize their clocks and hopping sequence with the primary. Note that a piconet can have only one primary station. The communication between the primary and secondary stations can be one-to-one or one-to-many. Figure 6.19 shows a piconet.

Figure 6.19 *Piconet*



Although a piconet can have a maximum of seven secondaries, additional secondaries can be in the *parked state*. A secondary in a parked state is synchronized with the primary, but cannot take part in communication until it is moved from the parked state to the active state. Because only eight stations can be active in a piconet, activating a station from the parked state means that an active station must go to the parked state.

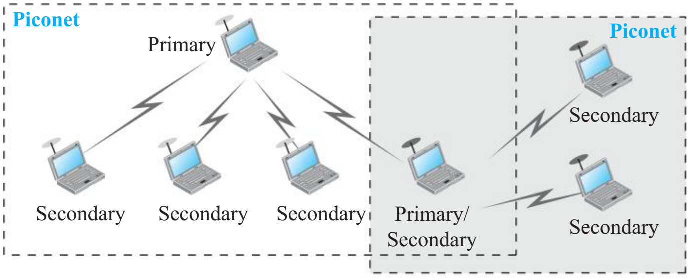
Scatternet

Piconets can be combined to form what is called a **scatternet.** A secondary station in one piconet can be the primary in another piconet. This station can receive messages from the primary in the first piconet (as a secondary) and, acting as a primary, deliver them to secondaries in the second piconet. A station can be a member of two piconets. Figure 6.20 illustrates a scatternet.

Bluetooth Devices

A Bluetooth device has a built-in short-range radio transmitter. The current data rate is 1 Mbps with a 2.4-GHz bandwidth. This means that there is a possibility of interference between the IEEE 802.11b wireless LANs and Bluetooth LANs.

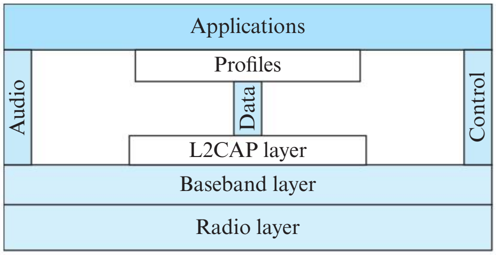
Figure 6.20 *Scatternet*



Bluetooth Layers

Bluetooth uses several layers that do not exactly match those of the Internet model we have defined in this book. Figure 6.21 shows these layers.

Figure 6.21 *Bluetooth layers*



L2CAP

The **Logical Link Control and Adaptation Protocol,** or **L2CAP** (L2 here means LL), is roughly equivalent to the LLC sublayer in LANs. It is used for data exchange on an ACL link; SCO channels do not use L2CAP. Figure 6.22 shows the format of the data packet at this level.

Figure 6.22 *L2CAP data packet format*

The 16-bit length field defines the size of the data, in bytes, coming from the upper layers. Data can be up to 65,535 bytes. The channel ID (CID) defines a unique identifier for the virtual channel created at this level (see below).

The L2CAP has specific duties: multiplexing, segmentation and reassembly, quality of service (QoS), and group management.

Multiplexing The L2CAP can do multiplexing. At the sender site, it accepts data from one of the upper-layer protocols, frames them, and delivers them to the baseband layer. At the receiver site, it accepts a frame from the baseband layer, extracts the data, and delivers them to the appropriate protocol layer. It creates a kind of virtual channel that we will discuss in later chapters on higher-level protocols.

Segmentation and Reassembly The maximum size of the payload field in the baseband layer is 2774 bits, or 343 bytes. This includes 4 bytes to define the packet and packet length. Therefore, the size of the packet that can arrive from an upper layer can only be 339 bytes. However, application layers sometimes need to send a data packet that can be up to 65,535 bytes (an Internet packet, for example). The L2CAP divides these large packets into segments and adds extra information to define the location of the segments in the original packet. The L2CAP segments the packet at the source and reassembles them at the destination.

QoS Bluetooth allows the stations to define a quality-of-service level. We discuss quality of service in Chapter 8. For the moment, it is sufficient to know that if no quality-of-service level is defined, Bluetooth defaults to what is called *best-effort* service; it will do its best under the circumstances.

Group Management Another functionality of L2CAP is to allow devices to create a type of logical addressing between themselves. This is similar to multicasting. For example, two or three secondary devices can be part of a multicast group to receive data from the primary.

Baseband Layer

The baseband layer is roughly equivalent to the MAC sublayer in LANs. The access method is TDMA (discussed later). The primary and secondary stations communicate with each other using time slots. The length of a time slot is exactly the same as the dwell time, 625 μs. This means that during the time that one frequency is used, a primary sends a frame to a secondary, or a secondary sends a frame to the primary. Note that the communication is only between the primary and a secondary; secondaries cannot communicate directly with one another.

TDMA Bluetooth uses a form of TDMA that is called **TDD-TDMA** (**time-division duplex TDMA**). TDD-TDMA is a kind of half-duplex communication in which the sender and receiver send and receive data, but not at the same time (half-duplex); however, the communication for each direction uses different hops. This is similar to walkie-talkies using different carrier frequencies.

- ***Single-Secondary Communication*** If the piconet has only one secondary, the TDMA operation is very simple. The time is divided into slots of 625 μs. The primary uses even-numbered slots (0, 2, 4, …); the secondary uses odd-numbered slots (1, 3, 5, …). TDD-TDMA allows the primary and the secondary to communicate in half-duplex mode. In slot 0, the primary sends and the secondary receives; in slot 1, the secondary sends and the primary receives. The cycle is repeated. Figure 6.23 shows the concept.
- ***Multiple-Secondary Communication*** The process is a little more involved if there is more than one secondary in the piconet. Again, the primary uses the even-numbered slots, but a secondary sends in the next odd-numbered slot if the packet in the previous slot was addressed to it. All secondaries listen on even-numbered slots, but only one secondary sends in any odd-numbered slot. Figure 6.24 shows a scenario.

Figure 6.23 *Single-secondary communication*

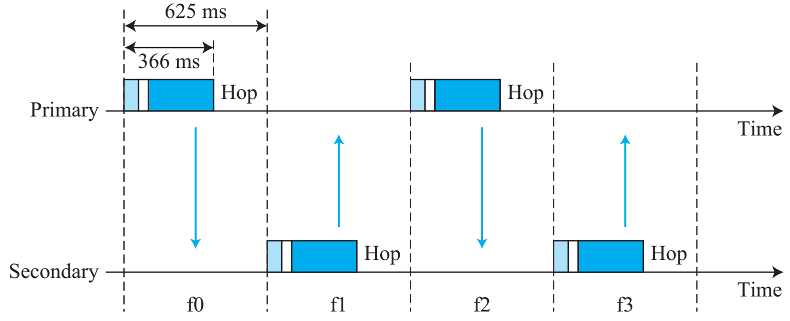
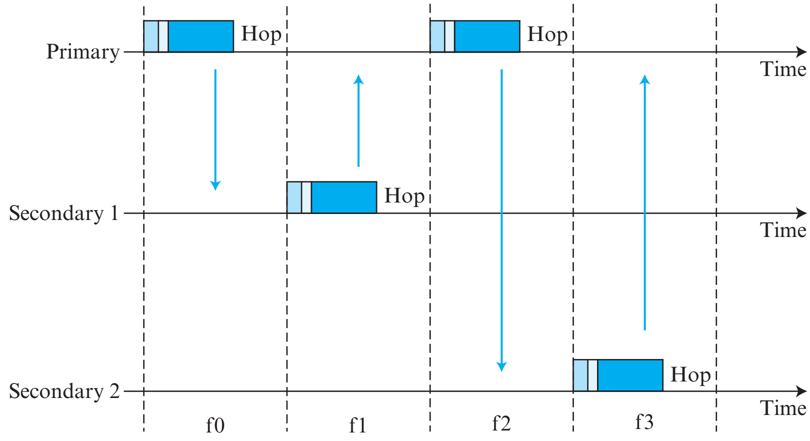


Figure 6.24 *Multiple-secondary communication*



Let us elaborate on the figure.

1. In slot 0, the primary sends a frame to secondary 1.
2. In slot 1, only secondary 1 sends a frame to the primary because the previous frame was addressed to secondary 1; other secondaries are silent.
3. In slot 2, the primary sends a frame to secondary 2.
4. In slot 3, only secondary 2 sends a frame to the primary because the previous frame was addressed to secondary 2; other secondaries are silent.
5. The cycle continues.

We can say that this access method is similar to a poll/select operation with reservations. When the primary selects a secondary, it also polls it. The next time slot is reserved for the polled station to send its frame. If the polled secondary has no frame to send, the channel is silent.

Links Two types of links can be created between a primary and a secondary: SCO links and ACL links.

- ***SCO*** A **synchronous connection-oriented (SCO)** link is used when avoiding latency (delay in data delivery) is more important than integrity (error-free delivery). In an SCO link, a physical link is created between the primary and a secondary by reserving specific slots at regular intervals. The basic unit of connection is two slots, one for each direction. If a packet is damaged, it is never retransmitted. SCO is used for real-time audio where avoiding delay is all-important. A secondary can create up to three SCO links with the primary, sending digitized audio (PCM) at 64 kbps in each link.
- ***ACL*** An **asynchronous connectionless link (ACL)** is used when data integrity is more important than avoiding latency. In this type of link, if a payload encapsulated in the frame is corrupted, it is retransmitted. A secondary returns an ACL frame in the available odd-numbered slot if the previous slot has been addressed to it. ACL can use one, three, or more slots and can achieve a maximum data rate of 721 kbps.

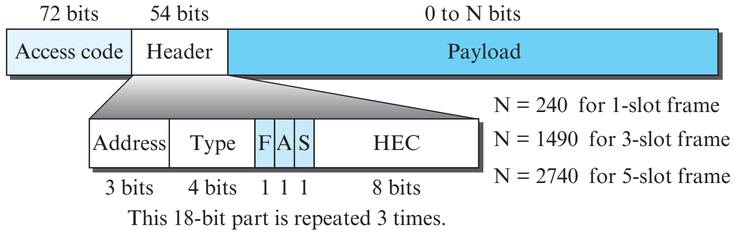
Frame Format A frame in the baseband layer can be one of three types: one-slot, three-slot, or five-slot. A slot, as we said before, is 625 μs. However, in a one-slot frame exchange, 259 μs is needed for hopping and control mechanisms. This means that a one-slot frame can last only 625 − 259, or 366 μs. With a 1-MHz bandwidth and 1 bit/Hz, the size of a one-slot frame is 366 bits.

A three-slot frame occupies three slots. However, since 259 μs is used for hopping, the length of the frame is $3 \times 625 - 259 = 1616$ μs or 1616 bits. A device that uses a three-slot frame remains at the same hop (at the same carrier frequency) for three slots. Even though only one hop number is used, three hop numbers are consumed. That means the hop number for each frame is equal to the first slot of the frame.

A five-slot frame also uses 259 bits for hopping, which means that the length of the frame is $5 \times 625 - 259 = 2866$ bits.

Figure 6.25 shows the format of the three frame types.

Figure 6.25 *Frame format types*



The following describes each field:

- ***Access code.*** This 72-bit field normally contains synchronization bits and the identifier of the primary to distinguish the frame of one piconet from another.
- ***Header.*** This 54-bit field is a repeated 18-bit pattern. Each pattern has the following subfields:

a. ***Address.*** The 3-bit address subfield can define up to seven secondaries (1 to 7). If the address is zero, it is used for broadcast communication from the primary to all secondaries.
b. ***Type.*** The 4-bit type subfield defines the type of data coming from the upper layers. We discuss these types later.
c. ***F.*** This 1-bit subfield is for flow control. When set (1), it indicates that the device is unable to receive more frames (buffer is full).
d. ***A.*** This 1-bit subfield is for acknowledgment. Bluetooth uses Stop-and-Wait ARQ; 1 bit is sufficient for acknowledgment.
e. ***S.*** This 1-bit subfield holds a sequence number. Bluetooth uses Stop-and-Wait ARQ; 1 bit is sufficient for sequence numbering.
f. ***HEC.*** The 8-bit header error correction subfield is a checksum to detect errors in each 18-bit header section. The header has three identical 18-bit sections. The receiver compares these three sections, bit by bit. If each of the corresponding bits is the same, the bit is accepted; if not, the majority opinion rules. This is a form of forward error correction (for the header only). This double error control is needed because the nature of the communication, via air, is very noisy. Note that there is no retransmission in this sublayer.

- ***Payload.*** This subfield can be 0 to 2740 bits long. It contains data or control information coming from the upper layers.

Radio Layer

The radio layer is roughly equivalent to the physical layer of the Internet model. Bluetooth devices are low-power and have a range of 10 m.

Band Bluetooth uses a 2.4-GHz ISM band divided into 79 channels of 1 MHz each.

FHSS Bluetooth uses the **frequency-hopping spread spectrum (FHSS)** method in the physical layer to avoid interference from other devices or other networks. Bluetooth hops 1600 times per second, which means that each device changes its modulation frequency 1600 times per second. A device uses a frequency for only 625 μs (1/1600 s) before it hops to another frequency; the dwell time is 625 μs.

Modulation To transform bits to a signal, Bluetooth uses a sophisticated version of FSK, called GFSK (FSK with Gaussian bandwidth filtering; a discussion of this topic is beyond the scope of this book). GFSK has a carrier frequency. Bit 1 is represented by a frequency deviation above the carrier; bit 0 is represented by a frequency deviation below the carrier. The frequencies, in megahertz, are defined according to the following formula for each channel.

$$f_c = 2402 + n \text{ MHz} \qquad n = 0, 1, 2, 3, \ldots, 78$$

For example, the first channel uses carrier frequency 2402 MHz (2.402 GHz), and the second channel uses carrier frequency 2403 MHz (2.403 GHz).

6.1.4 WiMAX

Worldwide Interoperability for Microwave Access (WiMAX) is an IEEE standard 802.16 (for fixed wireless) and 802.16e (for mobile wireless) that aims to provide the "last mile" broadband wireless access alternative to cable modem, telephone DSL service.

WiMAX offers optimum range and throughput to *line-of-sight (LOS)* subscribers (across a path that is not obstructed) and acceptable range and throughput to subscribers who are near or *non-line-of-sight (NLOS)* to the base station.

Many users compare WiMAX to WiFi. WiMAX, like WiFi, has a base station infrastructure, but it offers much more than WiFi. While WiFi only covers a range of about 100 yards, WiMAX covers 6 miles. WiMAX offers substantially greater security, reliability, quality of service, and throughput than WiFi.

Architecture

We briefly discuss the architecture of WiMAX in this section.

Base Station

The basic units of a WiMAX base station are radio and antenna. Each WiMax radio has both a transmitter and a receiver and transmits signals at a frequency between 2 and 11 GHz. WiMAX uses a Software Defined Radios (SDR) system.

Three different types of antennas (omni-directional, sector, and panel) are used in WiMax to optimize performance for a given application. WiMAX uses a beamsteering **adaptive antenna system** (**AAS**). While transmitting, an AAS antenna can focus its transmission energy in the direction of a receiver; while receiving, it can focus in the direction of the transmitting device.

Other interference avoidance measures used in WiMAX are the application of OFDMA, and MIMO antenna system. The OFDMA is a multiple-access method that allows simultaneous transmissions to and from several users and works well with AAS and MIMO to significantly improve throughput, increase link range, and reduce interference.

Subscriber Stations

The **customer premises equipment** (**CPE**), also called *subscriber unit*, is available in both indoor or outdoor versions. The indoor unit is the size of a cable or DSL modem and is self-installed, but because of radio losses, it requires the subscriber to be closer to a base station. The outdoor version is the size of a residential satellite dish and must be professionally installed.

Portable Unit

With the potential of mobile WiMax, there is an increasing focus on portable units, which include handsets, PC peripherals, and embedded devices in laptops, and consumer electronic devices (such as game terminals and MP3 players and the like).

Data-Link Layer

The MAC in WiFi uses contention access. This can cause subscriber stations distant from the AP to be repeatedly interrupted by closer stations. The MAC in WiMAX uses a scheduling algorithm. The subscriber station needs to compete only once for initial entry into the network. The access slot is then allocated to the subscriber.

Physical Layer

The 802.16e-2005 specifies 2 to 11 GHz range, **scalable OFDMA** (**SOFDMA**), MIMO antenna, and capability for full mobility support.

Application

WiMAX aims to provide cost-effective alternatives to several existing telecommunications infrastructures including the telephone company's copper wire and cellular networks and cable TV's coaxial cable infrastructure.

6.2 OTHER WIRELESS NETWORKS

In this section, we concentrate on other wireless networks. We first discuss cellular telephony, which is ubiquitous. We then talk about satellite networks. Before we discuss the above-mentioned wireless networks, let us discuss one access method that we postponed from Chapter 5: channelization, which is used in cellular and other wireless networks.

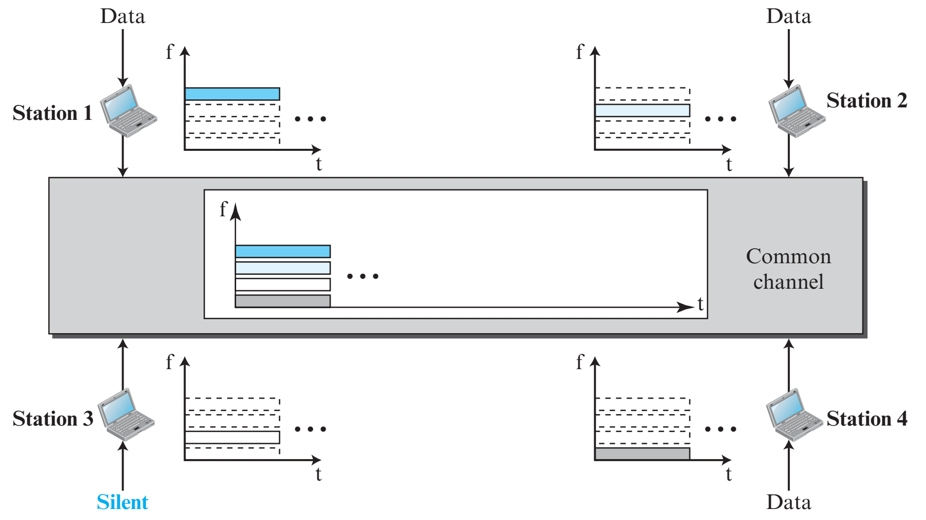
6.2.1 Channelization

Channelization (or *channel partition,* as it is sometime called) is a multiple-access method in which the available bandwidth of a link is shared in time, frequency, or through code, between different stations. In this section, we discuss three channelization protocols: FDMA, TDMA, and CDMA.

Frequency-Division Multiple Access (FDMA)

In **frequency-division multiple access (FDMA),** the available bandwidth is divided into frequency bands. Each station is allocated a band to send its data. In other words, each band is reserved for a specific station, and it belongs to the station all the time. Each station also uses a band-pass filter to confine the transmitter frequencies. To prevent station interferences, the allocated bands are separated from one another by small *guard bands*. Figure 6.26 shows the idea of FDMA.

Figure 6.26 *Frequency-division multiple access (FDMA)*



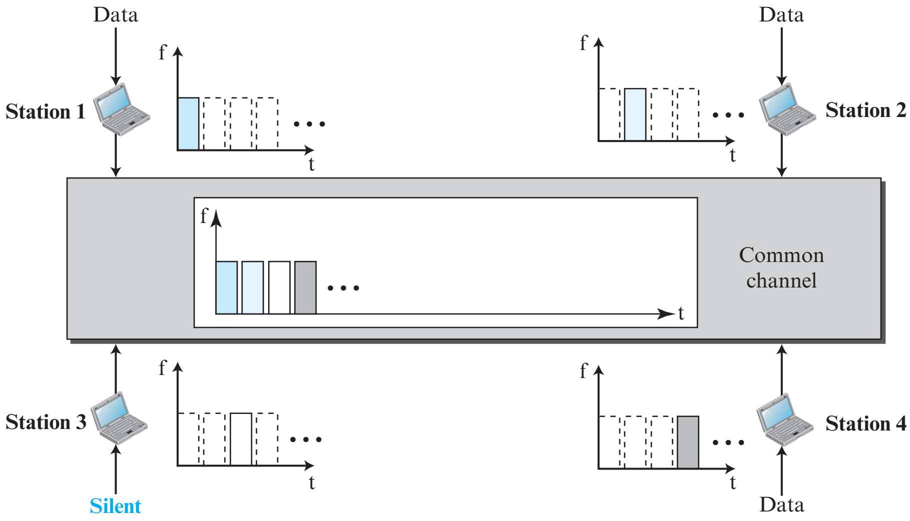
FDMA specifies a predetermined frequency band for the entire period of communication. This means that stream data (a continuous flow of data that may not be packetized) can easily be used with FDMA. We will see later how this feature can be used in cellular telephone systems.

We need to emphasize that FDMA is a link-layer protocol; it should not be confused with a multiplexing process, frequency-division multiplexing (FDM), which we discuss in Chapter 7. FDM is actually the implementation of FDMA at the physical layer.

Time-Division Multiple Access (TDMA)

In **time-division multiple access (TDMA),** the stations share the bandwidth of the channel in time. Each station is allocated a time slot during which it can send data. Each station transmits its data in its assigned time slot. Figure 6.27 shows the idea behind TDMA.

Figure 6.27 *Time-division multiple access (TDMA)*



The main problem with TDMA lies in achieving synchronization between the different stations. Each station needs to know the beginning of its slot and the location of its slot. This may be difficult because of propagation delays introduced in the system if the stations are spread over a large area. To compensate for the delays, we can insert *guard times*. Synchronization is normally accomplished by having some synchronization bits (normally referred to as preamble bits) at the beginning of each slot.

We also need to emphasize that although TDMA and TDM (time division multiplexing) conceptually seem the same, there are differences between them. TDM, as we see in Chapter 7, is a physical-layer technique that combines the data from slower channels and transmits them by using a faster channel. The process uses a physical multiplexer that interleaves data units from each channel.

TDMA, on the other hand, is an access method in the data-link layer. The data-link layer in each station tells its physical layer to use the allocated time slot. There is no physical multiplexer at the data-link layer.

Code-Division Multiple Access (CDMA)

Code-division multiple access (CDMA) was conceived several decades ago. Recent advances in electronic technology have finally made its implementation possible. CDMA differs from FDMA because only one channel occupies the entire bandwidth of the link. It differs from TDMA because all stations can send data simultaneously; there is no time-sharing.

Analogy

Let us first give an analogy. CDMA simply means communication with different codes. For example, in a large room with many people, two people can talk in English if nobody else understands English. Another two people can talk in Chinese if they are the only ones who understand Chinese, and so on. In other words, the common channel, the space of the room in this case, can easily allow communication between several couples, but in different languages (codes).

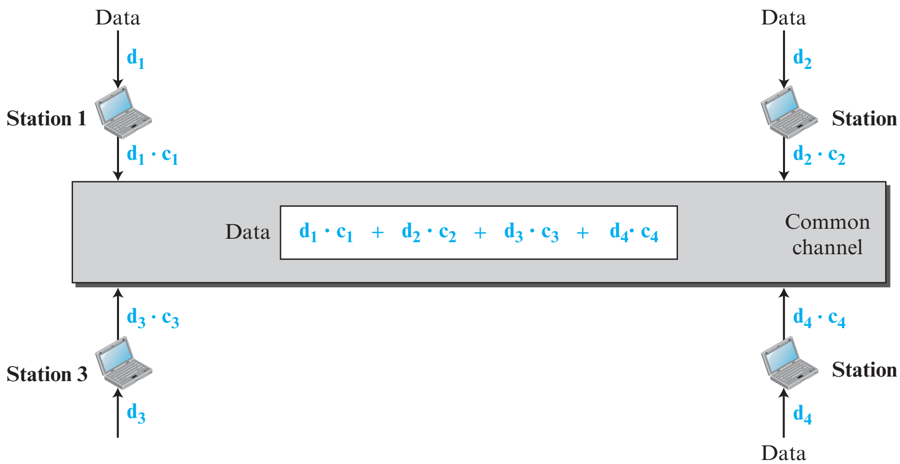
Idea

Let us assume we have four stations, 1, 2, 3, and 4, connected to the same channel. The data from station 1 are d_1, those from station 2 are d_2, and so on. The code assigned to the first station is c_1, to the second is c_2, and so on. We assume that the assigned codes have two properties.

1. If we multiply each code by another, we get 0.
2. If we multiply each code by itself, we get 4 (the number of stations).

With these two properties in mind, let us see how the above four stations can send data using the same common channel, as shown in Figure 6.28.

Figure 6.28 *Simple idea of communication with code*



Station 1 multiplies (a special kind of multiplication, as we will see) its data by its code to get $d_1 \cdot c_1$. Station 2 multiplies its data by its code to get $d_2 \cdot c_2$, and so on. The data that go on the channel are the sum of all these terms, as shown in the box. Any station that wants to receive data from one of the other three stations multiplies the data on the channel by the code of the sender. For example, suppose stations 1 and 2 are talking to each other. Station 2 wants to hear what station 1 is saying. It multiplies the data on the channel by c_1, the code of station 1.

Because $(c_1 \cdot c_1)$ is 4, but $(c_2 \cdot c_1)$, $(c_3 \cdot c_1)$, and $(c_4 \cdot c_1)$ are all 0s, station 2 divides the result by 4 to get the data from station 1.

$$\begin{aligned} \text{data} &= [(d_1 \cdot c_1 + d_2 \cdot c_2 + d_3 \cdot c_3 + d_4 \cdot c_4) \cdot c_1] / 4 \\ &= [d_1 \cdot c_1 \cdot c_1 + d_2 \cdot c_2 \cdot c_1 + d_3 \cdot c_3 \cdot c_1 + d_4 \cdot c_4 \cdot c_1] / 4 = (4 \times d_1) / 4 = d_1 \end{aligned}$$

Chips

CDMA is based on coding theory. Each station is assigned a code, which is a sequence of numbers called chips. Figure 6.29 shows the chips for the previous example.

Figure 6.29 *Chip sequences*

C_1	C_2	C_3	C_4
[+1 +1 +1 +1]	[+1 –1 +1 –1]	[+1 +1 –1 –1]	[+1 –1 –1 +1]

Later in this chapter we show how the codes are selected. For now, we need to know that we did not randomly choose the sequences; they were carefully selected. They are called **orthogonal sequences** and have the following properties:

1. Each sequence is made of N elements, where N is the number of stations and needs to be the power of 2.
2. If we multiply a sequence by a number, every element in the sequence is multiplied by that number. This is called multiplication of a sequence by a scalar. For example,

$$\mathbf{2 \cdot [+1\ +1\ -1\ -1] = [+2\ +2\ -2\ -2].}$$

3. If we multiply two equal sequences, element by element, and add the results, we get N, where N is the number of elements in each sequence. This is called the *inner product* of two equal sequences. For example,

$$\mathbf{[+1\ +1\ -1\ -1] \cdot [+1\ +1\ -1\ -1] = 1 + 1 + 1 + 1 = 4.}$$

4. If we multiply two different sequences, element by element, and add the results, we get 0. This is called the inner product of two different sequences. For example,

$$\mathbf{[+1\ +1\ -1\ -1] \cdot [+1\ +1\ +1\ +1] = 1 + 1 - 1 - 1 = 0.}$$

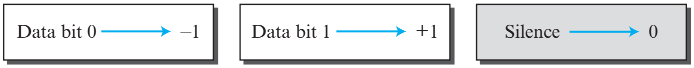
5. Adding two sequences means adding the corresponding elements. The result is another sequence. For example,

$$\mathbf{[+1\ +1\ -1\ -1] + [+1\ +1\ +1\ +1] = [+2\ +2\ 0\ 0].}$$

Data Representation

We follow some rules for encoding: If a station needs to send the bit 0, it encodes it as −1; if it needs to send the bit 1, it encodes it as +1. When a station is idle, it sends no signal, which is interpreted as a 0. These rules are shown in Figure 6.30.

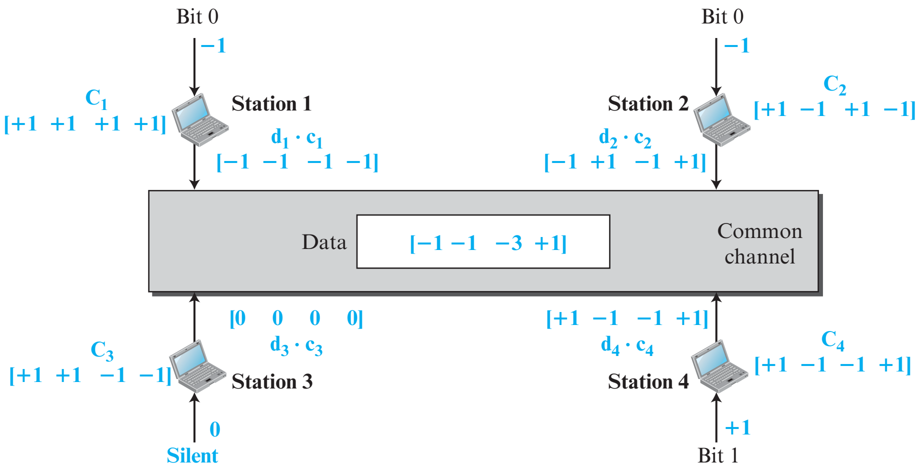
Figure 6.30 *Data representation in CDMA*



Encoding and Decoding

As a simple example, we show how four stations share the link during a 1-bit interval. The procedure can easily be repeated for additional intervals. We assume that stations 1 and 2 are sending a 0 bit and station 4 is sending a 1 bit. Station 3 is silent. The data at the sender site are translated to −1, −1, 0, and +1. Each station multiplies the corresponding number by its chip (its orthogonal sequence), which is unique for each station. The result is a new sequence which is sent to the channel. For simplicity, we assume that all stations send the resulting sequences at the same time. The sequence on the channel is the sum of all four sequences as defined before. Figure 6.31 shows the situation.

Figure 6.31 *Sharing channel in CDMA*



Now imagine station 3, which we said is silent, is listening to station 2. Station 3 multiplies the total data on the channel by the code for station 2, which is [+1 −1 +1 −1], to get

$$[-1\ -1\ -3\ +1] \cdot [+1\ -1\ +1\ -1] = -4$$
$$-4/4 = -1 \rightarrow \text{bit 0.}$$

Signal Level

The process can be better understood if we show the digital signal produced by each station and the data recovered at the destination (see Figure 6.32). The figure shows the corresponding signals for each station using NRZ-L signal (see Chapter 7) and the signal that is on the common channel.

Figure 6.32 *Digital signal created by four stations in CDMA*

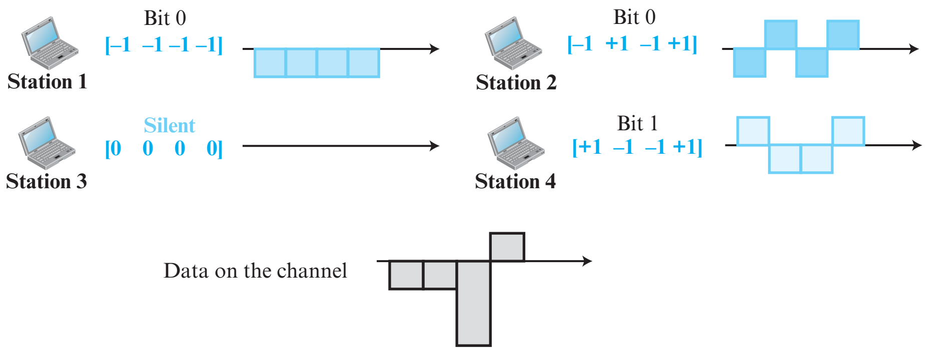
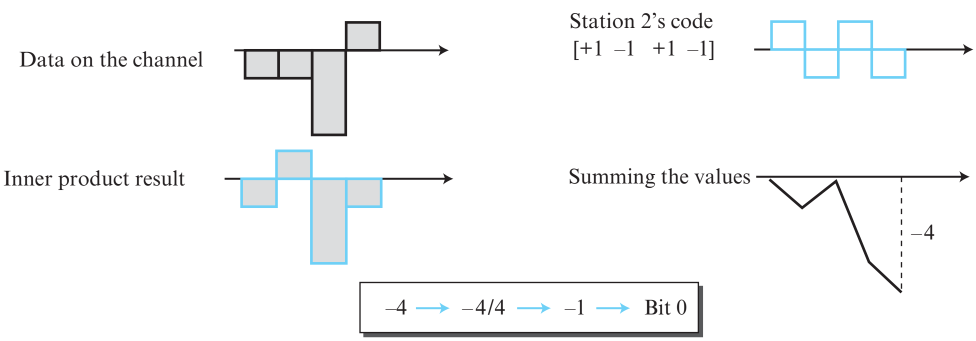


Figure 6.33 shows how station 3 can detect the data sent by station 2 by using the code for station 2. The total data on the channel are multiplied (inner product operation) by the signal representing the station 2 chip code to get a new signal. The station then integrates and adds the area under the signal, to get the value −4, which is divided by 4 and interpreted as bit 0.

Figure 6.33 *Decoding of the composite signal for one in CDMA*



Sequence Generation

To generate chip sequences, we use a **Walsh table,** which is a two-dimensional table with an equal number of rows and columns, as shown in Figure 6.34.

In the Walsh table, each row is a sequence of chips. W_1 for a one-chip sequence has one row and one column. We can choose −1 or +1 for the chip for this trivial table

Figure 6.34 *General rule and examples of creating Walsh tables*

$$W_1 = \begin{bmatrix} +1 \end{bmatrix} \qquad W_{2N} = \begin{bmatrix} W_N & W_N \\ W_N & \overline{W_N} \end{bmatrix}$$

a. Two basic rules

$$W_2 = \begin{bmatrix} +1 & +1 \\ +1 & -1 \end{bmatrix} \qquad W_4 = \begin{bmatrix} +1 & +1 & +1 & +1 \\ +1 & -1 & +1 & -1 \\ +1 & +1 & -1 & -1 \\ +1 & -1 & -1 & +1 \end{bmatrix}$$

b. Generation of W_1, W_2, and W_4

(we chose +1). According to Walsh, if we know the table for N sequences W_N, we can create the table for $2N$ sequences W_{2N}, as shown in Figure 6.34. The W_N with the overbar W_N stands for the complement of W_N, where each +1 is changed to −1 and vice versa. Figure 6.34 also shows how we can create W_2 and W_4 from W_1. After we select W_1, W_2 can be made from four W_1's, with the last one the complement of W_1. After W_2 is generated, W_4 can be made of four W_2's, with the last one the complement of W_2. Of course, W_8 is composed of four W_4's, and so on. Note that after W_N is made, each station is assigned a chip corresponding to a row.

Something we need to emphasize is that the number of sequences N needs to be a power of 2. In other words, we need to have $N = 2^m$.

Example 6.1

Find the chips for a network with

a. Two stations

b. Four stations

Solution

We can use the rows of W_2 and W_4 in Figure 6.34:

a. For a two-station network, we have [+1 +1] and [+1 −1].

b. For a four-station network we have [+1 +1 +1 +1], [+1 −1 +1 −1], [+1 +1 −1 −1], and [+1 −1 −1 +1].

Example 6.2

What is the number of sequences if we have 90 stations in our network?

Solution

The number of sequences needs to be 2^m. We need to choose $m = 7$ and $N = 2^7$ or 128. We can then use 90 of the sequences as the chips.

Example 6.3

Prove that a receiving station can get the data sent by a specific sender if it multiplies the entire data on the channel by the sender's chip code and then divides it by the number of stations.

Solution

Let us prove this for the first station, using our previous four-station example. We can say that the data on the channel $D = (d_1 \cdot c_1 + d_2 \cdot c_2 + d_3 \cdot c_3 + d_4 \cdot c_4)$. The receiver, which wants to get the data sent by station 1, multiplies these data by c_1.

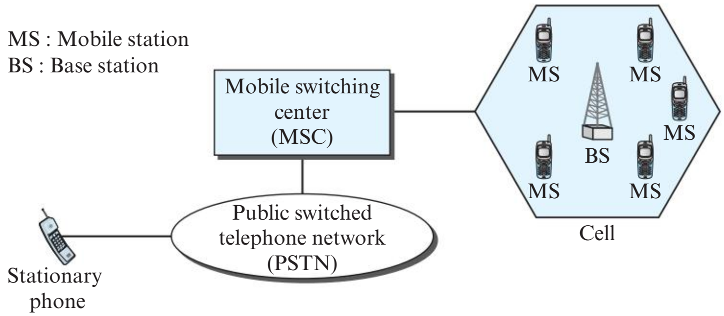
$$\begin{aligned}
[\mathbf{D} \cdot \mathbf{c}_1] / 4 &= [(\mathbf{d}_1 \cdot \mathbf{c}_1 + \mathbf{d}_2 \cdot \mathbf{c}_2 + \mathbf{d}_3 \cdot \mathbf{c}_3 + \mathbf{d}_4 \cdot \mathbf{c}_4) \cdot \mathbf{c}_1] / 4 \\
&= [\mathbf{d}_1 \cdot \mathbf{c}_1 \cdot \mathbf{c}_1 + \mathbf{d}_2 \cdot \mathbf{c}_2 \cdot \mathbf{c}_1 + \mathbf{d}_3 \cdot \mathbf{c}_3 \cdot \mathbf{c}_1 + \mathbf{d}_4 \cdot \mathbf{c}_4 \cdot \mathbf{c}_1] / 4 \\
&= [\mathbf{d}_1 \times 4 + \mathbf{d}_2 \times 0 + \mathbf{d}_3 \times 0 + \mathbf{d}_4 \times 0] / 4 = [\mathbf{d}_1 \times 4] / 4 = \mathbf{d}_1
\end{aligned}$$

6.2.2 Cellular Telephony

Cellular telephony is designed to provide communications between two moving units, called *mobile stations* (MSs), or between one mobile unit and one stationary unit, often called a *land unit*. A service provider must be able to locate and track a caller, assign a channel to the call, and transfer the channel from base station to base station as the caller moves out of range.

To make this tracking possible, each cellular service area is divided into small regions called *cells*. Each cell contains an antenna and is controlled by a solar- or AC-powered network station, called the *base station* (BS). Each base station, in turn, is controlled by a switching office, called a **mobile switching center (MSC).** The MSC coordinates communication between all the base stations and the telephone central office. It is a computerized center that is responsible for connecting calls, recording call information, and billing (see Figure 6.35).

Figure 6.35 *Cellular system*



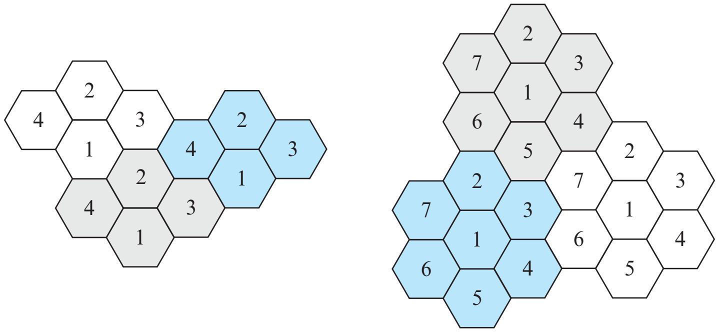
Cell size is not fixed and can be increased or decreased depending on the population of the area. The typical radius of a cell is 1 to 12 mi. High-density areas require more, geographically smaller cells to meet traffic demands than do low-density areas. Once determined, cell size is optimized to prevent the interference of adjacent cell signals. The transmission power of each cell is kept low to prevent its signal from interfering with those of other cells.

Frequency-Reuse Principle

In general, neighboring cells cannot use the same set of frequencies for communication because it may create interference for the users located near the cell boundaries.

However, the set of frequencies available is limited, and frequencies need to be reused. A frequency reuse pattern is a configuration of *N* cells, *N* being the **reuse factor,** in which each cell uses a unique set of frequencies. When the pattern is repeated, the frequencies can be reused. There are several different patterns. Figure 6.36 shows two of them.

Figure 6.36 *Frequency reuse patterns*



a. Reuse factor of 4 b. Reuse factor of 7

The numbers in the cells define the pattern. The cells with the same number in a pattern can use the same set of frequencies. We call these cells the *reusing cells*. As Figure 6.36 shows, in a pattern with reuse factor 4, only one cell separates the cells using the same set of frequencies. In the pattern with reuse factor 7, two cells separate the reusing cells.

Transmitting

To place a call from a mobile station, the caller enters a code of 7 or 10 digits (a phone number) and presses the send button. The mobile station then scans the band, seeking a setup channel with a strong signal, and sends the data (phone number) to the closest base station using that channel. The base station relays the data to the MSC. The MSC sends the data on to the telephone central office. If the called party is available, a connection is made and the result is relayed back to the MSC. At this point, the MSC assigns an unused voice channel to the call, and a connection is established. The mobile station automatically adjusts its tuning to the new channel, and communication can begin.

Receiving

When a mobile phone is called, the telephone central office sends the number to the MSC. The MSC searches for the location of the mobile station by sending query signals to each cell in a process called *paging*. Once the mobile station is found, the MSC transmits a ringing signal and, when the mobile station answers, assigns a voice channel to the call, allowing voice communication to begin.

Handoff

It may happen that, during a conversation, the mobile station moves from one cell to another. When it does, the signal may become weak. To solve this problem, the MSC monitors the level of the signal every few seconds. If the strength of the signal diminishes, the MSC seeks a new cell that can better accommodate the communication. The MSC then changes the channel carrying the call (hands the signal off from the old channel to a new one).

Hard Handoff Early systems used a hard **handoff.** In a hard handoff, a mobile station only communicates with one base station. When the MS moves from one cell to another, communication must first be broken with the previous base station before communication can be established with the new one. This may create a rough transition.

Soft Handoff New systems use a soft handoff. In this case, a mobile station can communicate with two base stations at the same time. This means that, during handoff, a mobile station may continue with the new base station before breaking off from the old one.

Roaming

One feature of cellular telephony is called **roaming.** Roaming means, in principle, that a user can have access to communication or can be reached where there is coverage. A service provider usually has limited coverage. Neighboring service providers can provide extended coverage through a roaming contract. The situation is similar to snail mail between countries. The charge for delivery of a letter between two countries can be divided upon agreement by the two countries.

First Generation (1G)

Cellular telephony is now in its fourth generation. The first generation was designed for voice communication using analog signals. We discuss one first-generation mobile system used in North America, AMPS.

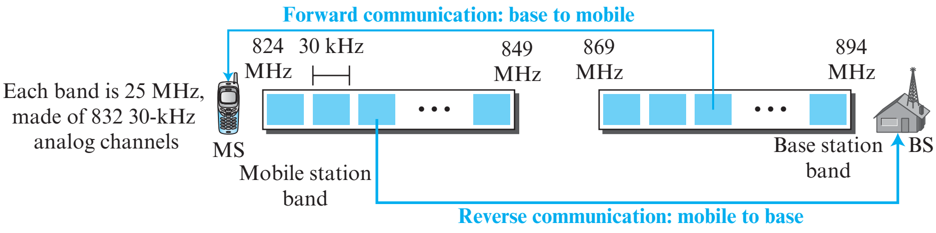
AMPS

Advanced Mobile Phone System (AMPS) is one of the leading analog cellular systems in North America. It uses FDMA (see Chapter 5) to separate channels in a link.

AMPS is an analog cellular phone system using FDMA.

Bands AMPS operates in the ISM 800-MHz band. The system uses two separate analog channels, one for forward (base station to mobile station) communication and one for reverse (mobile station to base station) communication. The band between 824 and 849 MHz carries reverse communication; the band between 869 and 894 MHz carries forward communication (see Figure 6.37).

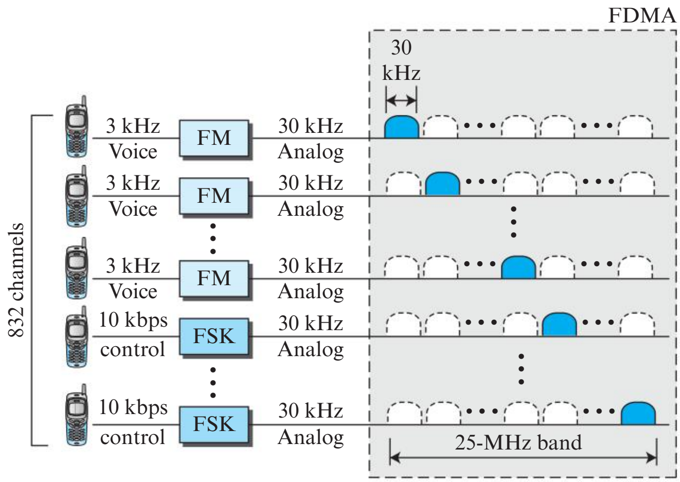
Figure 6.37 *Cellular bands for AMPS*



Each band is divided into 832 channels. However, two providers can share an area, which means 416 channels in each cell for each provider. Out of these 416, 21 channels are used for control, which leaves 395 channels. AMPS has a frequency reuse factor of 7; this means only one-seventh of these 395 traffic channels are actually available in a cell.

Transmission AMPS uses FM and FSK for modulation. Figure 6.38 shows the transmission in the reverse direction. Voice channels are modulated using FM, and control channels use FSK to create 30-kHz analog signals. AMPS uses FDMA to divide each 25-MHz band into 30-kHz channels.

Figure 6.38 *AMPS reverse communication band*



Second Generation (2G)

To provide higher-quality (less noise-prone) mobile voice communications, the second generation of the cellular phone network was developed. While the first generation was designed for analog voice communication, the second generation was mainly designed for digitized voice. Three major systems evolved in the second generation: D-AMPS, GSM, and CDMA.

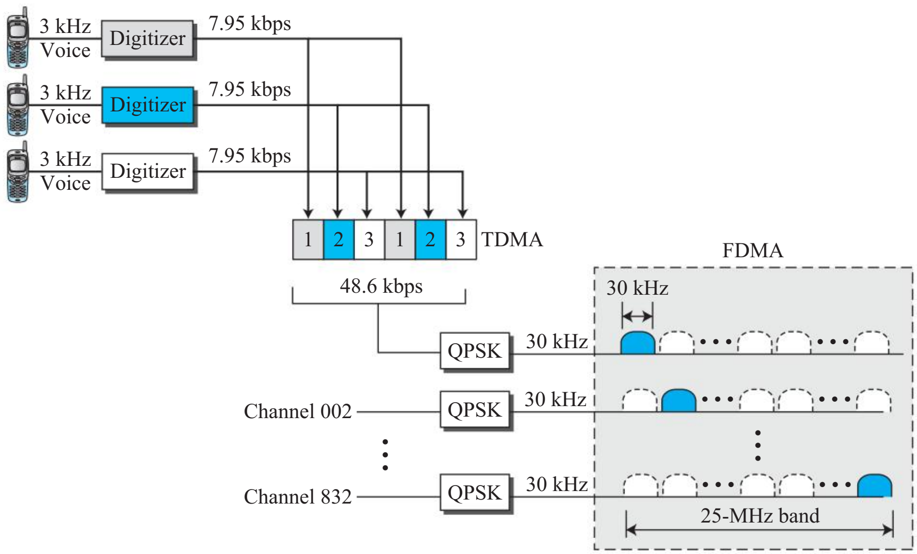
D-AMPS

The product of the evolution of the analog AMPS into a digital system is **digital AMPS (D-AMPS).** D-AMPS was designed to be backward-compatible with AMPS. This means that in a cell, one telephone can use AMPS and another D-AMPS. D-AMPS was first defined by IS-54 (Interim Standard 54) and later revised by IS-136.

Band D-AMPS uses the same bands and channels as AMPS.

Transmission Each voice channel is digitized using a very complex PCM and compression technique. A voice channel is digitized to 7.95 kbps. Three 7.95-kbps digital voice channels are combined using TDMA. The result is 48.6 kbps of digital data; much of this is overhead. As Figure 6.39 shows, the system sends 25 frames per second, with

Figure 6.39 *D-AMPS*



1944 bits per frame. Each frame lasts 40 ms (1/25) and is divided into six slots shared by three digital channels; each channel is allotted two slots.

Each slot holds 324 bits. However, only 159 bits come from the digitized voice; 64 bits are for control and 101 bits are for error correction. In other words, each channel drops 159 bits of data into each of the two channels assigned to it. The system adds 64 control bits and 101 error-correcting bits.

The resulting 48.6 kbps of digital data modulates a carrier using QPSK; the result is a 30-kHz analog signal. Finally, the 30-kHz analog signals share a 25-MHz band (FDMA). D-AMPS has a frequency reuse factor of 7.

D-AMPS, or IS-136, is a digital cellular phone system using TDMA and FDMA.

GSM

The **Global System for Mobile Communication (GSM)** is a European standard that was developed to provide a common second-generation technology for all Europe. The aim was to replace a number of incompatible first-generation technologies.

Bands GSM uses two bands for duplex communication. Each band is 25 MHz in width, shifted toward 900 MHz, as shown in Figure 6.40. Each band is divided into 124 channels of 200 kHz separated by guard bands.

Transmission Figure 6.41 shows a GSM system. Each voice channel is digitized and compressed to a 13-kbps digital signal. Each slot carries 156.25 bits. Eight slots share a frame (TDMA). Twenty-six frames also share a multiframe (TDMA). We can calculate the bit rate of each channel as follows.

$$\textbf{Channel data rate} = (1/120\text{ ms}) \times 26 \times 8 \times 156.25 = 270.8\text{ kbps}$$

Figure 6.40 *GSM bands*

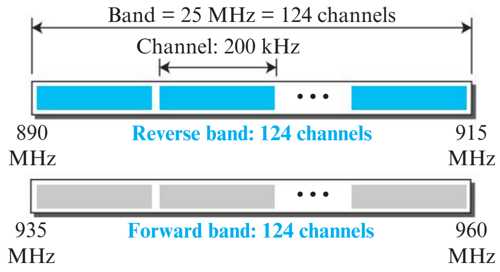
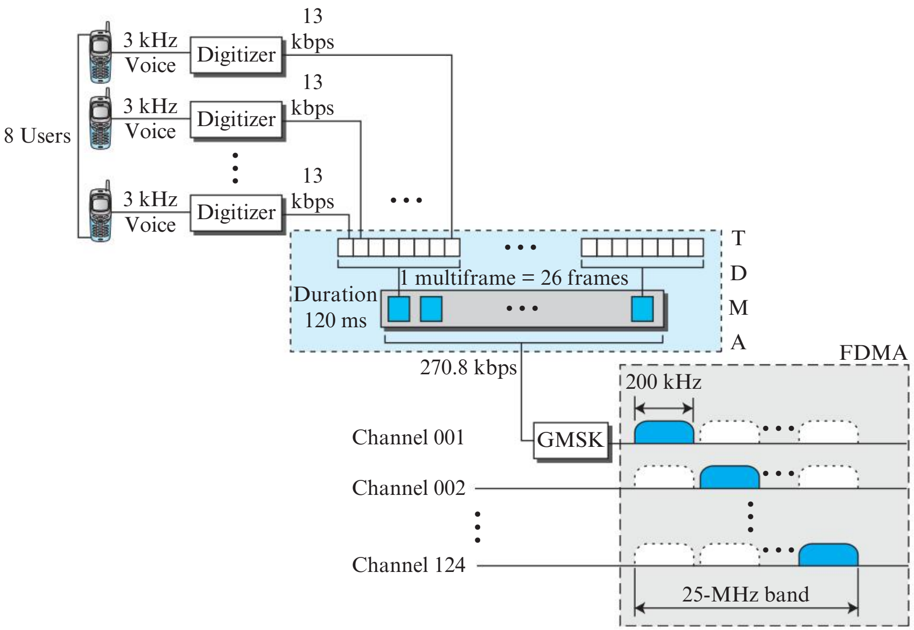


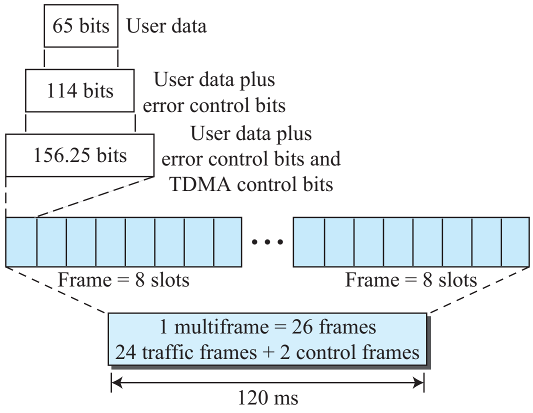
Figure 6.41 *GSM*



Each 270.8-kbps digital channel modulates a carrier using GMSK (a form of FSK used mainly in European systems); the result is a 200-kHz analog signal. Finally 124 analog channels of 200 kHz are combined using FDMA. The result is a 25-MHz band. Figure 6.42 shows the user data and overhead in a multiframe.

The reader may have noticed the large amount of overhead in TDMA. The user data are only 65 bits per slot. The system adds extra bits for error correction to make it 114 bits per slot. To this, control bits are added to bring it up to 156.25 bits per slot. Eight slots are encapsulated in a frame. Twenty-four traffic frames and two additional control frames make a multiframe. A multiframe has a duration of 120 ms. However, the architecture does define superframes and hyperframes that do not add any overhead; we will not discuss them here.

Figure 6.42 *Multiframe components*



Reuse Factor Because of the complex error correction mechanism, GSM allows a reuse factor as low as 3.

GSM is a digital cellular phone system using TDMA and FDMA.

IS-95

One of the dominant second-generation standards in North America is **Interim Standard 95 (IS-95).** It is based on CDMA and DSSS.

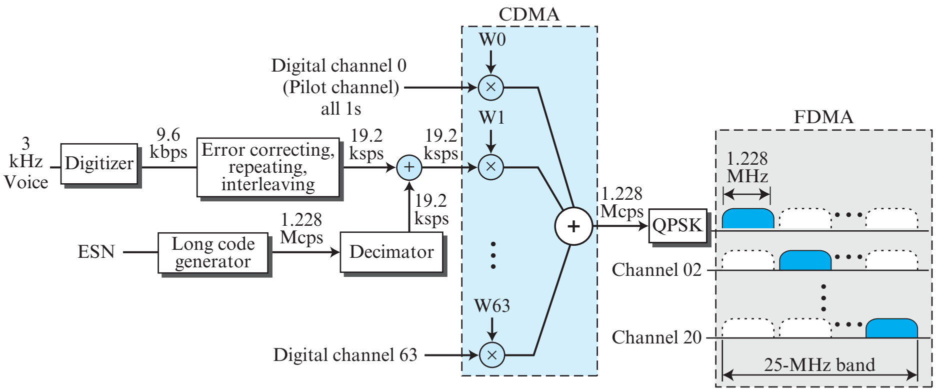
Bands and Channels IS-95 uses two bands for duplex communication. The bands can be the traditional ISM 800-MHz band or the ISM 1900-MHz band. Each band is divided into 20 channels of 1.228 MHz separated by guard bands. Each service provider is allotted 10 channels. IS-95 can be used in parallel with AMPS. Each IS-95 channel is equivalent to 41 AMPS channels (41×30 kHz = 1.23 MHz).

Synchronization All base channels need to be synchronized to use CDMA. To provide synchronization, bases use the services of GPS (Global Positioning System), a satellite system that we discuss in the next section.

Forward Transmission IS-95 has two different transmission techniques: one for use in the forward (base to mobile) direction and another for use in the reverse (mobile to base) direction. In the forward direction, communications between the base and all mobiles are synchronized; the base sends synchronized data to all mobiles. Figure 6.43 shows a simplified diagram for the forward direction.

Each voice channel is digitized, producing data at a basic rate of 9.6 kbps. After adding error-correcting and repeating bits, and interleaving, the result is a signal of 19.2 ksps (kilo signals per second). This output is now scrambled using a 19.2-ksps signal. The scrambling signal is produced from a long code generator that uses the electronic serial number (ESN) of the mobile station and generates 2^{42} pseudorandom chips, each chip having 42 bits. Note that the chips are generated pseudorandomly, not randomly, because the pattern repeats itself. The output of the long code generator is

Figure 6.43 *IS-95 forward transmission*



fed to a decimator, which chooses 1 bit out of 64 bits. The output of the decimator is used for scrambling. The scrambling is used to create privacy; the ESN is unique for each station.

The result of the scrambler is combined using CDMA. For each traffic channel, one Walsh 64 × 64 row chip is selected. The result is a signal of 1.228 Mcps (megachips per second).

19.2 ksps × 64 cps = 1.228 Mcps

The signal is fed into a QPSK modulator to produce a signal of 1.228 MHz. The resulting bandwidth is shifted appropriately, using FDMA. An analog channel creates 64 digital channels, of which 55 channels are traffic channels (carrying digitized voice). Nine channels are used for control and synchronization:

a. Channel 0 is a pilot channel. This channel sends a continuous stream of 1s to mobile stations. The stream provides bit synchronization, serves as a phase reference for demodulation, and allows the mobile station to compare the signal strength of neighboring bases for handoff decisions.

b. Channel 32 gives information about the system to the mobile station.

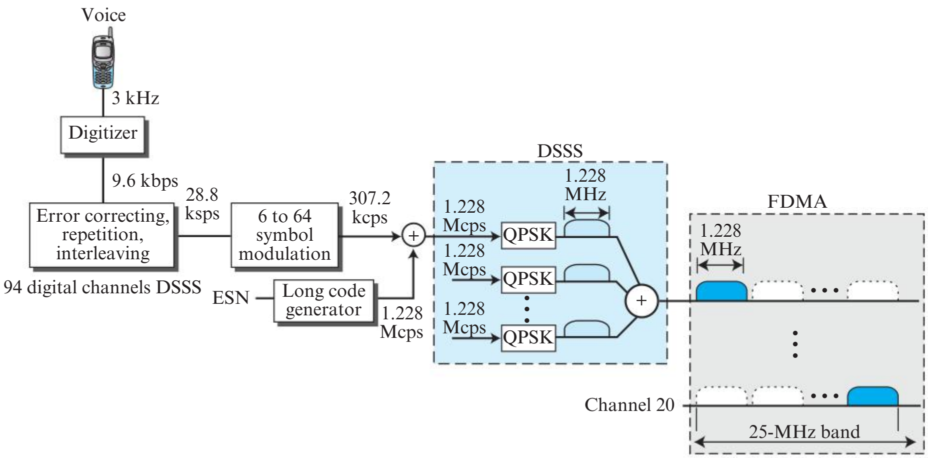
c. Channels 1 to 7 are used for paging, to send messages to one or more mobile stations.

d. Channels 8 to 31 and 33 to 63 are traffic channels carrying digitized voice from the base station to the corresponding mobile station.

Reverse Transmission The use of CDMA in the forward direction is possible because the pilot channel sends a continuous sequence of 1s to synchronize transmission. The synchronization is not used in the reverse direction because we need an entity to do that, which is not feasible. Instead of CDMA, the reverse channels use DSSS (direct sequence spread spectrum), which we discuss in Chapter 7. Figure 6.44 shows a simplified diagram for reverse transmission.

Each voice channel is digitized, producing data at a rate of 9.6 kbps. However, after adding error-correcting and repeating bits, plus interleaving, the result is a signal

Figure 6.44 *IS-95 reverse transmission*



of 28.8 ksps. The output is now passed through a 6/64 symbol modulator. The symbols are divided into six-symbol chunks, and each chunk is interpreted as a binary number (from 0 to 63). The binary number is used as the index to a 64×64 Walsh matrix for selection of a row of chips. Note that this procedure is not CDMA; each bit is not multiplied by the chips in a row. Each six-symbol chunk is replaced by a 64-chip code. This is done to provide a kind of orthogonality; it differentiates the streams of chips from the different mobile stations. The result creates a signal of 307.2 kcps or $(28.8/6) \times 64$.

Spreading is the next step; each chip is spread into 4. Again the ESN of the mobile station creates a long code of 42 bits at a rate of 1.228 Mcps, which is 4 times 307.2. After spreading, each signal is modulated using QPSK, which is slightly different from the one used in the forward direction; we do not go into details here. Note that there is no multiple-access mechanism here; all reverse channels send their analog signal into the air, but the correct chips will be received by the base station due to spreading.

Although we can create $2^{42} - 1$ digital channels in the reverse direction (because of the long code generator), normally 94 channels are used; 62 are traffic channels, and 32 are channels used to gain access to the base station.

IS-95 is a digital cellular phone system using CDMA/DSSS and FDMA.

Two Data Rate Sets IS-95 defines two data rate sets, with four different rates in each set. The first set defines 9600, 4800, 2400, and 1200 bps. If, for example, the selected rate is 1200 bps, each bit is repeated 8 times to provide a rate of 9600 bps. The second set defines 14,400, 7200, 3600, and 1800 bps. This is possible by reducing the number of bits used for error correction. The bit rates in a set are related to the activity of the channel. If the channel is silent, only 1200 bits can be transferred, which improves the spreading by repeating each bit 8 times.

Frequency-Reuse Factor In an IS-95 system, the frequency-reuse factor is normally 1 because the interference from neighboring cells cannot affect CDMA or DSSS transmission.

Soft Handoff Every base station continuously broadcasts signals using its pilot channel. This means a mobile station can detect the pilot signal from its cell and neighboring cells. This enables a mobile station to do a soft handoff in contrast to a hard handoff.

Third Generation (3G)

The third generation of cellular telephony refers to a combination of technologies that provide both digital data and voice communication. Using a small portable device, a person is able to talk to anyone else in the world with a voice quality similar to that of the existing fixed telephone network. A person can download and watch a movie, download and listen to music, surf the Internet or play games, have a video conference, and do much more. One of the interesting characteristics of a third-generation system is that the portable device is always connected; you do not need to dial a number to connect to the Internet.

The third-generation concept started in 1992, when ITU issued a blueprint called the **Internet Mobile Communication 2000 (IMT-2000).** The blueprint defines some criteria for third-generation technology as outlined below:

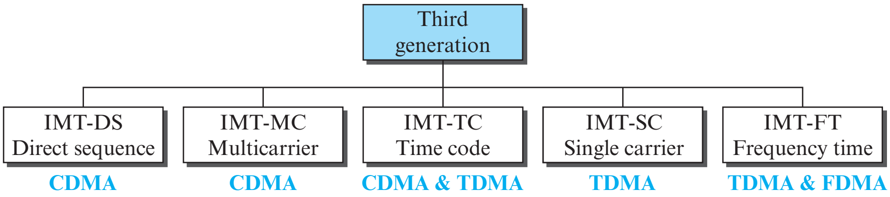
a. Voice quality comparable to that of the existing public telephone network.
b. Data rate of 144 kbps for access in a moving vehicle (car), 384 kbps for access as the user walks (pedestrians), and 2 Mbps for the stationary user (office or home).
c. Support for packet-switched and circuit-switched data services.
d. A band of 2 GHz.
e. Bandwidths of 2 MHz.
f. Interface to the Internet.

The main goal of third-generation cellular telephony is to provide universal personal communication.

IMT-2000 Radio Interface

Figure 6.45 shows the radio interfaces (wireless standards) adopted by IMT-2000. All five are developed from second-generation technologies. The first two evolve from CDMA technology. The third evolves from a combination of CDMA and TDMA. The fourth evolves from TDMA, and the last evolves from both FDMA and TDMA.

Figure 6.45 *IMT-2000 radio interfaces*



IMT-DS This approach uses a version of CDMA called wideband CDMA or W-CDMA. W-CDMA uses a 5-MHz bandwidth. It was developed in Europe, and it is compatible with the CDMA used in IS-95.

IMT-MC This approach was developed in North America and is known as CDMA 2000. It is an evolution of CDMA technology used in IS-95 channels. It combines the new wideband (15-MHz) spread spectrum with the narrowband (1.25-MHz) CDMA of IS-95. It is backward-compatible with IS-95. It allows communication on multiple 1.25-MHz channels (1, 3, 6, 9, 12 times), up to 15 MHz. The use of the wider channels allows it to reach the 2-Mbps data rate defined for the third generation.

IMT-TC This standard uses a combination of W-CDMA and TDMA. The standard tries to reach the IMT-2000 goals by adding TDMA multiplexing to W-CDMA.

IMT-SC This standard uses only TDMA.

IMT-FT This standard uses a combination of FDMA and TDMA.

Fourth Generation (4G)

The fourth generation of cellular telephony is expected to be a complete evolution in wireless communications. Some of the objectives defined by the 4G working group are as follows:

a. A spectrally efficient system.
b. High network capacity.
c. Data rate of 100 Mbit/s for access in a moving car and 1 Gbit/s for stationary users.
d. Data rate of at least 100 Mbit/s between any two points in the world.
e. Smooth handoff across heterogeneous networks.
f. Seamless connectivity and global roaming across multiple networks.
g. High quality of service for next generation multimedia support (quality of service will be discussed in Chapter 8).
h. Interoperability with existing wireless standards.
i. All IP, packet-switched network.

The fourth generation is only packet-based (unlike 3G) and supports IPv6. This provides better multicast, security, and route optimization capabilities.

Access Scheme

To increase efficiency, capacity, and scalability, new access techniques are being considered for 4G. For example, **orthogonal FDMA** (**OFDMA**) and **interleaved FDMA** (**IFDMA**) are being considered respectively for the downlink and uplink of the next generation **Universal Mobile Telecommunications System** (**UMTS**). Similarly, **multi-carrier code division multiple access** (**MC-CDMA**) is proposed for the IEEE 802.20 standard.

Modulation

More efficient quadrature amplitude modulation (64-QAM) is being proposed for use with the Long Term Evolution (LTE) standards.

Radio System

The fourth generation uses a **Software Defined Radio** (**SDR**) system. Unlike a common radio which uses hardware, the components of an SDR are pieces of software and thus felexible. The SDR can change its program to shift its frequencies to mitigate frequency interference.

Antenna

The **multiple-input multiple-output** (**MIMO**) and **multi-user MIMO** (**MU-MIMO**) antenna system, a branch of intelligent antenna, is proposed for 4G. Using this antenna system together with special multiplexing, 4G allows independent streams to be transmitted simultaneously from all the antennas to increase the data rate into multiple folds. MIMO also allows the transmitter and receiver coordinates to move to an open frequency when interference occurs.

Applications

At the present rates of 15-30 Mbit/s, 4G is capable of providing users with streaming high-definition television. At rates of 100 Mbit/s, the content of a DVD-5 can be downloaded within about 5 minutes for offline access.

6.2.3 Satellite Networks

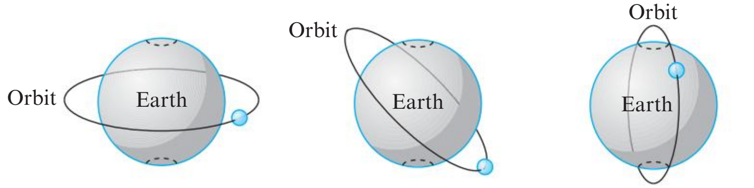
A *satellite network* is a combination of nodes, some of which are satellites, that provides communication from one point on the Earth to another. A node in the network can be a satellite, an Earth station, or an end-user terminal or telephone. Although a natural satellite, such as the moon, can be used as a relaying node in the network, the use of artificial satellites is preferred because we can install electronic equipment on the satellite to regenerate the signal that has lost its energy during travel. Another restriction on using natural satellites is their distances from the Earth, which create a long delay in communication.

Satellite networks are like cellular networks in that they divide the planet into cells. Satellites can provide transmission capability to and from any location on Earth, no matter how remote. This advantage makes high-quality communication available to undeveloped parts of the world without requiring a huge investment in ground-based infrastructure.

Orbits

An artificial satellite needs to have an *orbit,* the path in which it travels around the Earth. The orbit can be equatorial, inclined, or polar, as shown in Figure 6.46.

Figure 6.46 *Satellite orbits*



a. Equatorial-orbit satellite b. Inclined-orbit satellite c. Polar-orbit satellite

The period of a satellite, the time required for a satellite to make a complete trip around the Earth, is determined by Kepler's law, which defines the period as a function of the distance of the satellite from the center of the Earth.

Example 6.4

What is the period of the moon, according to Kepler's law?

$$\textbf{Period} = C \times \textbf{distance}^{1.5}$$

Here C is a constant approximately equal to 1/100. The period is in seconds and the distance in kilometers.

Solution

The moon is located approximately 384,000 km above the Earth. The radius of the Earth is 6378 km. Applying the formula, we get the following.

$$\textbf{Period} = (1/100) \times (384{,}000 + 6378)^{1.5} = 2{,}439{,}090 \text{ s} = 1 \text{ month}$$

Example 6.5

According to Kepler's law, what is the period of a satellite that is located at an orbit approximately 35,786 km above the Earth?

Solution

Applying the formula, we get the following.

$$\textbf{Period} = (1/100) \times (35{,}786 + 6378)^{1.5} = 86{,}579 \text{ s} = 24 \text{ h}$$

This means that a satellite located at 35,786 km has a period of 24 h, which is the same as the rotation period of the Earth. A satellite like this is said to be *stationary* to the Earth. The orbit, as we will see, is called a *geostationary orbit*.

Footprint

Satellites process microwaves with bidirectional antennas (line-of-sight). Therefore, the signal from a satellite is normally aimed at a specific area called the **footprint.** The signal power at the center of the footprint is maximum. The power decreases as we move out from the footprint center. The boundary of the footprint is the location where the power level is at a predefined threshold.

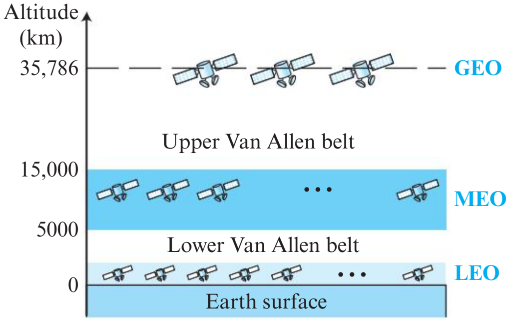
Three Categories of Satellites

Based on the location of the orbit, satellites can be divided into three categories: **geostationary Earth orbit (GEO), low-Earth-orbit (LEO),** and **medium-Earth-orbit (MEO).**

Figure 6.47 shows the satellite altitudes with respect to the surface of the Earth. There is only one orbit, at an altitude of 35,786 km for the GEO satellite. MEO satellites are located at altitudes between 5000 and 15,000 km. LEO satellites are normally below an altitude of 2000 km.

One reason for having different orbits is the existence of two Van Allen belts. A Van Allen belt is a layer that contains charged particles. A satellite orbiting in one of these two belts would be totally destroyed by the energetic charged particles. The MEO orbits are located between these two belts.

Figure 6.47 *Satellite orbit altitudes*



Frequency Bands for Satellite Communication

The frequencies reserved for satellite microwave communication are in the gigahertz (GHz) range. Each satellite sends and receives over two different bands. Transmission from the Earth to the satellite is called the *uplink*. Transmission from the satellite to the Earth is called the *downlink*. Table 6.5 gives the band names and frequencies for each range.

Table 6.5 *Satellite frequency bands*

Band	*Downlink, GHz*	*Uplink, GHz*	*Bandwidth, MHz*
L	1.5	1.6	15
S	1.9	2.2	70
C	4.0	6.0	500
Ku	11.0	14.0	500
Ka	20.0	30.0	3500

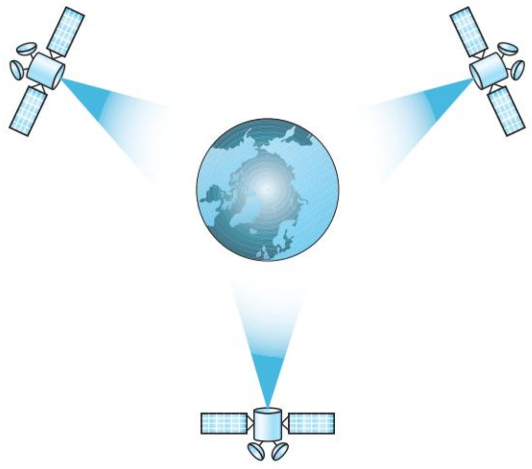
GEO Satellites

Line-of-sight propagation requires that the sending and receiving antennas be locked onto each other's location at all times (one antenna must have the other in sight). For this reason, a satellite that moves faster or slower than the Earth's rotation is useful only for short periods. To ensure constant communication, the satellite must move at the same speed as the Earth so that it seems to remain fixed above a certain spot. Such satellites are called *geostationary*.

Because orbital speed is based on the distance from the planet, only one orbit can be geostationary. This orbit occurs at the equatorial plane and is approximately 22,000 mi from the surface of the Earth.

But one geostationary satellite cannot cover the whole Earth. One satellite in orbit has line-of-sight contact with a vast number of stations, but the curvature of the Earth still keeps much of the planet out of sight. It takes a minimum of three satellites equidistant from each other in geostationary Earth orbit (GEO) to provide full global transmission. Figure 6.48 shows three satellites, each 120° from another in geosynchronous orbit around the equator. The view is from the North Pole.

Figure 6.48 *Satellites in geostationary orbit*



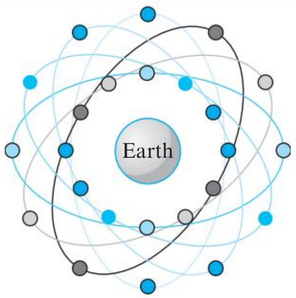
MEO Satellites

Medium-Earth-orbit (MEO) satellites are positioned between the two Van Allen belts. A satellite at this orbit takes approximately 6 to 8 hours to circle the Earth.

Global Positioning System

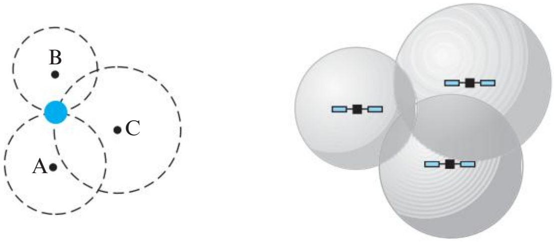
One example of a MEO satellite system is the **Global Positioning System (GPS),** contracted and operated by the U.S. Department of Defense, orbiting at an altitude about 18,000 km (11,000 mi) above the Earth. The system consists of 24 satellites and is used for land, sea, and air navigation to provide time and location for vehicles and ships. GPS uses 24 satellites in six orbits, as shown in Figure 6.49. The orbits and the locations of the satellites in each orbit are designed in such a way that, at any time, four satellites are visible from any point on Earth. A GPS receiver has an almanac that tells the current position of each satellite.

Figure 6.49 *Orbits for global positioning system (GPS) satellites*



Trilateration GPS is based on a principle called **trilateration.** The terms *trilateration* and *triangulation* are normally used interchangeably. We use the word *trilateration,* which means using three distances, instead of **triangulation,** which may mean using three angles. On a plane, if we know our distance from three points, we know exactly where we are. Let us say that we are 10 miles away from point A, 12 miles away from point B, and 15 miles away from point C. If we draw three circles with the centers at A, B, and C, we must be somewhere on circle A, somewhere on circle B, and somewhere on circle C. These three circles meet at one single point (if our distances are correct); this is our position. Figure 6.50a shows the concept.

Figure 6.50 *Trilateration on a plane*



a. Two-dimensional trilateration b. Three-dimensional trilateration

In three-dimensional space, the situation is different. Three spheres meet in two points, as shown in Figure 6.50b. We need at least four spheres to find our exact position in space (longitude, latitude, and altitude). However, if we have additional facts about our location (for example, we know that we are not inside the ocean or somewhere in space), three spheres are enough, because one of the two points, where the spheres meet, is so improbable that the other can be selected without a doubt.

Measuring the distance The trilateration principle can find our location on the Earth if we know our distance from three satellites and know the position of each satellite. The position of each satellite can be calculated by a GPS receiver (using the predetermined path of the satellites). The GPS receiver, then, needs to find its distance from at least three GPS satellites (center of the spheres). Measuring the distance is done using a principle called one-way ranging. For the moment, let us assume that all GPS satellites and the receiver on the Earth are synchronized. Each of 24 satellites synchronously transmits a complex signal, each satellite's signal having a unique pattern. The computer on the receiver measures the delay between the signals from the satellites and its copy of the signals to determine the distances to the satellites.

Synchronization The previous discussion was based on the assumption that the satellites' clock are synchronized with each other and with the receiver's clock. Satellites use atomic clocks, which are precise and can function synchronously with each other. The receiver's clock, however, is a normal quartz clock (an atomic clock costs more than $50,000), and there is no way to synchronize it with the satellite clocks. There is an unknown offset between the satellite clocks and the receiver clock that introduces a corresponding offset in the distance calculation. Because of this offset, the measured distance is called a *pseudorange*.

GPS uses an elegant solution to the clock offset problem, by recognizing that the offset's value is the same for all satellites being used. The calculation of position becomes finding four unknowns: the x_r, y_r, z_r coordinates of the receiver, and common clock offset dt. For finding these four unknown values, we need at least four equations. This means that we need to measure pseudoranges from four satellites instead of three. If we call the four measured pseudoranges PR_1, PR_2, PR_3, and PR_4 and the coordinates of each satellite x_i, y_i, and z_i (for $i = 1$ to 4), we can find the four previously mentioned unknown values using the following four equations (the four unknown values are shown in color).

$$\mathbf{PR_1} = [(x_1 - x_r)^2 + (y_1 - y_r)^2 + (z_1 - z_r)^2]^{1/2} + \mathbf{c} \times \mathbf{d}t$$
$$\mathbf{PR_2} = [(x_2 - x_r)^2 + (y_2 - y_r)^2 + (z_2 - z_r)^2]^{1/2} + \mathbf{c} \times \mathbf{d}t$$
$$\mathbf{PR_3} = [(x_3 - x_r)^2 + (y_3 - y_r)^2 + (z_3 - z_r)^2]^{1/2} + \mathbf{c} \times \mathbf{d}t$$
$$\mathbf{PR_4} = [(x_4 - x_r)^2 + (y_4 - y_r)^2 + (z_4 - z_r)^2]^{1/2} + \mathbf{c} \times \mathbf{d}t$$

The coordinates used in the above formulas are in an Earth-Centered Earth-Fixed (ECEF) reference frame, which means that the origin of the coordinate space is at the center of the Earth and the coordinate space rotates with the Earth. This implies that the ECEF coordinates of a fixed point on the surface of the earth do not change.

Application GPS is used by military forces. For example, thousands of portable GPS receivers were used during the Persian Gulf war by foot soldiers, vehicles, and helicopters. Another use of GPS is in navigation. The driver of a car can find the location of the car. The driver can then consult a database in the memory of the automobile to be directed to the destination. In other words, GPS gives the location of the car, and the database uses this information to find a path to the destination. A very interesting application is clock synchronization. As we mentioned previously, the IS-95 cellular telephone system uses GPS to create time synchronization between the base stations.

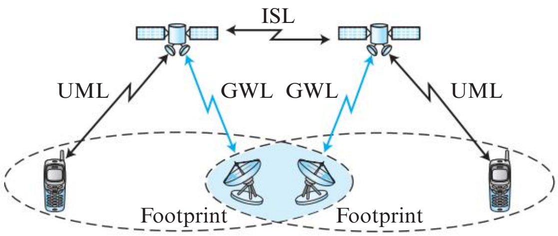
LEO Satellites

Low-Earth-orbit (LEO) satellites have polar orbits. The altitude is between 500 and 2000 km, with a rotation period of 90 to 120 min. The satellite has a speed of 20,000 to 25,000 km/h. A LEO system usually has a cellular type of access, similar to the cellular telephone system. The footprint normally has a diameter of 8000 km. Because LEO satellites are close to Earth, the round-trip time propagation delay is normally less than 20 ms, which is acceptable for audio communication.

A LEO system is made of a constellation of satellites that work together as a network; each satellite acts as a switch. Satellites that are close to each other are connected through intersatellite links (ISLs). A mobile system communicates with the satellite through a user mobile link (UML). A satellite can also communicate with an Earth station (gateway) through a gateway link (GWL). Figure 6.51 shows a typical LEO satellite network.

LEO satellites can be divided into three categories: little LEOs, big LEOs, and broadband LEOs. The little LEOs operate under 1 GHz. They are mostly used for low-data-rate messaging. The big LEOs operate between 1 and 3 GHz. **Globalstar** is one of the examples of a big LEO satellite system. It uses 48 satellites in 6 polar orbits with each orbit hosting 8 satellites. The orbits are located at an altitude of almost 1400 km. Iridium systems are also examples of big LEOs. The **Iridium** system has 66 satellites divided into 6 orbits, with 11 satellites in each orbit. The orbits are at an altitude of

Figure 6.51 *LEO satellite system*



750 km. The satellites in each orbit are separated from one another by approximately 32° of latitude. The broadband LEOs provide communication similar to fiber-optic networks. The first broadband LEO system was **Teledesic.** Teledesic is a system of satellites that provides fiber-optic-like communication (broadband channels, low error rate, and low delay). Its main purpose is to provide broadband Internet access for users all over the world. It is sometimes called "Internet in the sky." The project was started in 1990 by Craig McCaw and Bill Gates; later, other investors joined the consortium. The project is scheduled to be fully functional in the near future.

6.3 MOBILE IP

As mobile and personal computers such as notebooks become increasingly popular, we need to think about mobile IP, the extension of IP protocol that allows mobile computers to be connected to the Internet at any location where the connection is possible. In this section, we discuss this issue.

6.3.1 Addressing

The main problem that must be solved in providing mobile communication using the IP protocol is addressing.

Stationary Hosts

The original IP addressing was based on the assumption that a host is stationary, attached to one specific network. A router uses an IP address to route an IP datagram. As we learned in Chapter 5, an IP address has two parts: a prefix and a suffix. The prefix associates a host with a network. For example, the IP address 10.3.4.24/8 defines a host attached to the network 10.0.0.0/8. This implies that a host in the Internet does not have an address that it can carry with itself from one place to another. The address is valid only when the host is attached to the network. If the network changes, the address is no longer valid. Routers use this association to route a packet; they use the prefix to deliver the packet to the network to which the host is attached. This scheme works perfectly with **stationary hosts.**

The IP addresses are designed to work with stationary hosts because part of the address defines the network to which the host is attached.

Mobile Hosts

When a host moves from one network to another, the IP addressing structure needs to be modified. Several solutions have been proposed.

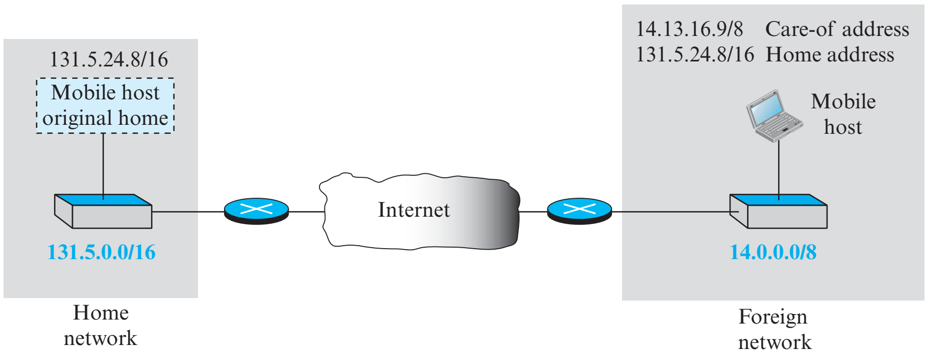
Changing the Address

One simple solution is to let the **mobile host** change its address as it goes to the new network. The host can use DHCP (see Chapter 4) to obtain a new address to associate it with the new network. This approach has several drawbacks. First, the configuration files would need to be changed. Second, each time the computer moves from one network to another, it must be rebooted. Third, the DNS tables (see Chapter 2) need to be revised so that every other host in the Internet is aware of the change. Fourth, if the host roams from one network to another during a transmission, the data exchange will be interrupted. This is because the ports and IP addresses of the client and the server must remain constant for the duration of the connection.

Two Addresses

The approach that is more feasible is the use of two addresses. The host has its original address, called the **home address,** and a temporary address, called the **care-of address.** The home address is permanent; it associates the host to its **home network,** the network that is the permanent home of the host. The care-of address is temporary. When a host moves from one network to another, the care-of address changes; it is associated with the **foreign network,** the network to which the host moves. Figure 6.52 shows the concept.

Figure 6.52 *Home address and care-of address*



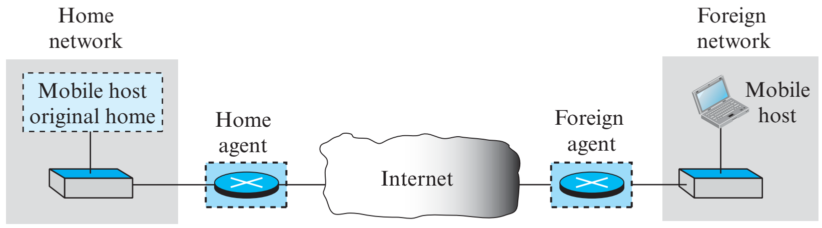
Mobile IP has two addresses for a mobile host: one home address and one care-of address. The home address is permanent; the care-of address changes as the mobile host moves from one network to another.

When a mobile host visits a foreign network, it receives its care-of address during the agent discovery and registration phase, described later.

6.3.2 Agents

To make the change of address transparent to the rest of the Internet requires a **home agent** and a **foreign agent.** Figure 6.53 shows the position of a home agent relative to the home network and a foreign agent relative to the foreign network.

Figure 6.53 *Home agent and foreign agent*



We have shown the home and the foreign agents as routers, but we need to emphasize that their specific function as an agent is performed in the application layer. In other words, they are both routers and hosts.

Home Agent

The home agent is usually a router attached to the home network of the mobile host. The home agent acts on behalf of the mobile host when a remote host sends a packet to the mobile host. The home agent receives the packet and sends it to the foreign agent.

Foreign Agent

The foreign agent is usually a router attached to the foreign network. The foreign agent receives and delivers packets sent by the home agent to the mobile host.

The mobile host can also act as a foreign agent. In other words, the mobile host and the foreign agent can be the same. However, to do this, a mobile host must be able to receive a care-of address by itself, which can be done through the use of DHCP. In addition, the mobile host needs the necessary software to allow it to communicate with the home agent and to have two addresses: its home address and its care-of address. This dual addressing must be transparent to the application programs.

When the mobile host acts as a foreign agent, the care-of address is called a **collocated care-of address.**

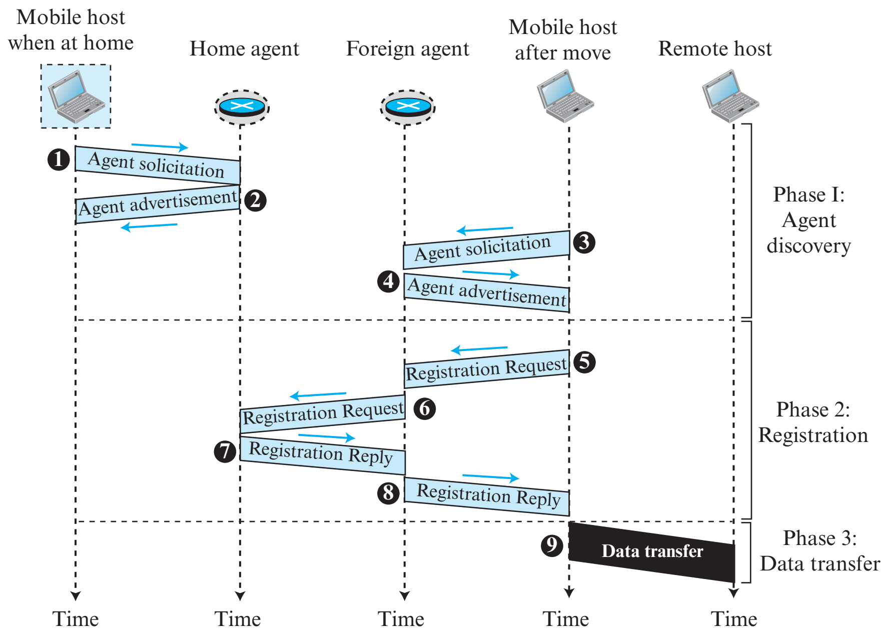
When the mobile host and the foreign agent are the same, the care-of address is called a collocated care-of address.

The advantage of using a collocated care-of address is that the mobile host can move to any network without worrying about the availability of a foreign agent. The disadvantage is that the mobile host needs extra software to act as its own foreign agent.

6.3.3 Three Phases

To communicate with a remote host, a mobile host goes through three phases: agent discovery, registration, and data transfer, as shown in Figure 6.54.

Figure 6.54 *Remote host and mobile host communication*



The first phase, agent discovery, involves the mobile host, the foreign agent, and the home agent. The second phase, registration, also involves the mobile host and the two agents. Finally, in the third phase, the remote host is also involved. We discuss each phase separately.

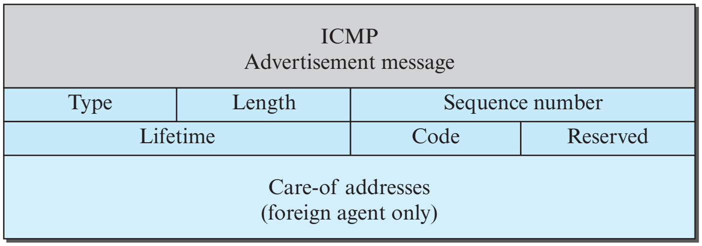
Agent Discovery

The first phase in mobile communication, *agent discovery,* consists of two subphases. A mobile host must discover (learn the address of) a home agent before it leaves its home network. A mobile host must also discover a foreign agent after it has moved to a foreign network. This discovery consists of learning the care-of address as well as the foreign agent's address. The discovery involves two types of messages: advertisement and solicitation.

Agent Advertisement

When a router advertises its presence on a network using an ICMP router advertisement, it can append an *agent advertisement* to the packet if it acts as an agent. Figure 6.55 shows how an agent advertisement is piggybacked to the router advertisement packet.

Figure 6.55 *Agent advertisement*



Mobile IP does not use a new packet type for agent advertisement; it uses the router advertisement packet of ICMP, and appends an agent advertisement message.

The field descriptions are as follows:

- ***Type.*** The 8-bit type field is set to 16.
- ***Length.*** The 8-bit length field defines the total length of the extension message (not the length of the ICMP advertisement message).
- ***Sequence number.*** The 16-bit sequence number field holds the message number. The recipient can use the sequence number to determine if a message is lost.
- ***Lifetime.*** The lifetime field defines the number of seconds that the agent will accept requests. If the value is a string of 1s, the lifetime is infinite.
- ***Code.*** The code field is an 8-bit flag in which each bit is set (1) or unset (0). The meanings of the bits are shown in Table 6.6.

Table 6.6 *Code Bits*

Bit	*Meaning*
0	Registration required. No collocated care-of address.
1	Agent is busy and does not accept registration at this moment.
2	Agent acts as a home agent.
3	Agent acts as a foreign agent.
4	Agent uses minimal encapsulation.
5	Agent uses generic routing encapsulation (GRE).
6	Agent supports header compression.
7	Unused (0).

- ***Care-of Addresses.*** This field contains a list of addresses available for use as care-of addresses. The mobile host can choose one of these addresses. The selection of this care-of address is announced in the registration request. Note that this field is used only by a foreign agent.

Agent Solicitation
When a mobile host has moved to a new network and has not received agent advertisements, it can initiate an *agent solicitation.* It can use the ICMP solicitation message to inform an agent that it needs assistance.

Mobile IP does not use a new packet type for agent solicitation; it uses the router solicitation packet of ICMP.

Registration

The second phase in mobile communication is *registration.* After a mobile host has moved to a foreign network and discovered the foreign agent, it must register. There are four aspects of registration:

1. The mobile host must register itself with the foreign agent.
2. The mobile host must register itself with its home agent. This is normally done by the foreign agent on behalf of the mobile host.
3. The mobile host must renew registration if it has expired.
4. The mobile host must cancel its registration (deregistration) when it returns home.

Request and Reply
To register with the foreign agent and the home agent, the mobile host uses a *registration request* and a registration reply as shown in Figure 6.54.

Registration Request A registration request is sent from the mobile host to the foreign agent to register its care-of address and also to announce its home address and home agent address. The foreign agent, after receiving and registering the request, relays the message to the home agent. Note that the home agent now knows the address of the foreign agent because the IP packet that is used for relaying has the IP address of the foreign agent as the source address. Figure 6.56 shows the format of the registration request.

Figure 6.56 *Registration request format*

<table>
<tr><td>Type</td><td>Flag</td><td colspan="2">Lifetime</td></tr>
<tr><td colspan="4">Home address</td></tr>
<tr><td colspan="4">Home agent address</td></tr>
<tr><td colspan="4">Care-of address</td></tr>
<tr><td colspan="4">Identification</td></tr>
<tr><td colspan="4">Extensions ...</td></tr>
</table>

The field descriptions are as follows:

- ***Type.*** The 8-bit type field defines the type of the message. For a request message the value of this field is 1.
- ***Flag.*** The 8-bit flag field defines forwarding information. The value of each bit can be set or unset. The meaning of each bit is given in Table 6.7.

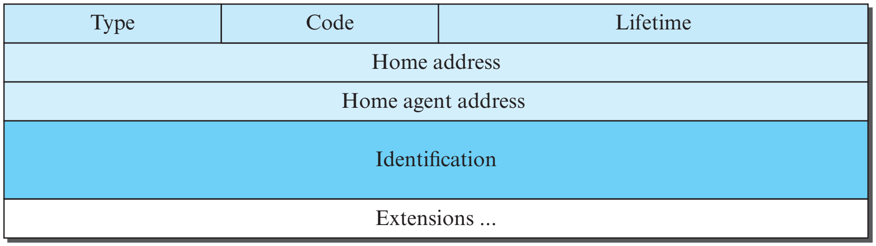
Table 6.7 *Registration request flag field bits*

Bit	*Meaning*
0	Mobile host requests that home agent retain its prior care-of address.
1	Mobile host requests that home agent tunnel any broadcast message.
2	Mobile host is using collocated care-of address.
3	Mobile host requests that home agent use minimal encapsulation.
4	Mobile host requests generic routing encapsulation (GRE).
5	Mobile host requests header compression.
6–7	Reserved bits.

- ***Lifetime.*** This field defines the number of seconds the registration is valid. If the field is a string of 0s, the request message is asking for deregistration. If the field is a string of 1s, the lifetime is infinite.
- ***Home address.*** This field contains the permanent (first) address of the mobile host.
- ***Home agent address.*** This field contains the address of the home agent.
- ***Care-of address.*** This field is the temporary (second) address of the mobile host.
- ***Identification.*** This field contains a 64-bit number that is inserted into the request by the mobile host and repeated in the reply message. It matches a request with a reply.
- ***Extensions.*** Variable length extensions are used for authentication. They allow a home agent to authenticate the mobile agent. We discuss authentication in Chapter 10.

Registration Reply A registration reply is sent from the home agent to the foreign agent and then relayed to the mobile host. The reply confirms or denies the registration request. Figure 6.57 shows the format of the registration reply.

Figure 6.57 *Registration reply format*



The fields are similar to those of the registration request with the following exceptions. The value of the type field is 3. The code field replaces the flag field and shows the result of the registration request (acceptance or denial). The care-of address field is not needed.

Encapsulation

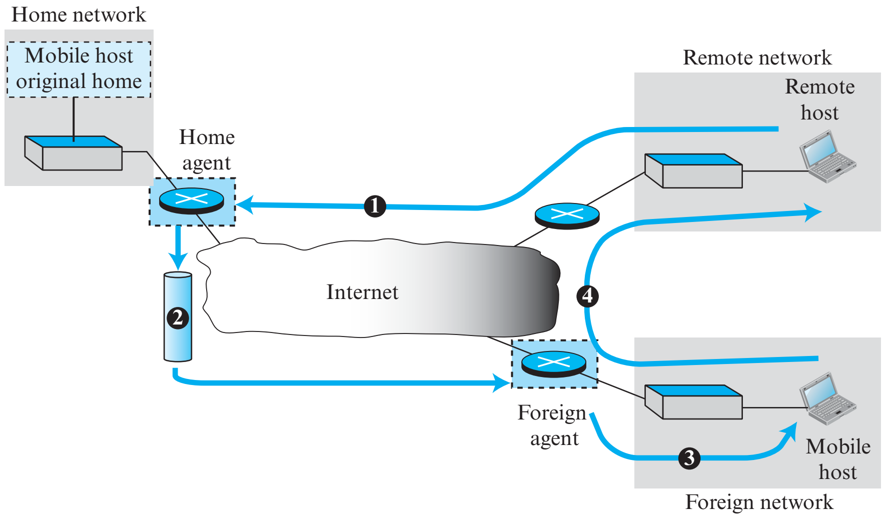
Registration messages are encapsulated in a UDP user datagram. An agent uses the well-known port 434; a mobile host uses an ephemeral port.

A registration request or reply is sent by UDP using the well-known port 434.

Data Transfer

After agent discovery and registration, a mobile host can communicate with a remote host. Figure 6.58 shows the idea.

Figure 6.58 *Data transfer*



From Remote Host to Home Agent

When a remote host wants to send a packet to the mobile host, it uses its address as the source address and the home address of the mobile host as the destination address. In other words, the remote host sends a packet as though the mobile host is at its home network. The packet, however, is intercepted by the home agent, which pretends it is the mobile host. This is done using the proxy ARP technique discussed in Chapter 5. Path 1 of Figure 6.58 shows this step.

From Home Agent to Foreign Agent

After receiving the packet, the home agent sends the packet to the foreign agent, using the tunneling concept discussed in Chapter 4. The home agent encapsulates the whole IP packet inside another IP packet using its address as the source and the foreign agent's address as the destination. Path 2 of Figure 6.58 shows this step.

From Foreign Agent to Mobile Host

When the foreign agent receives the packet, it removes the original packet. However, since the destination address is the home address of the mobile host, the foreign agent consults a registry table to find the care-of address of the mobile host. (Otherwise, the packet would just be sent back to the home network.) The packet is then sent to the care-of address. Path 3 of Figure 6.58 shows this step.

From Mobile Host to Remote Host

When a mobile host wants to send a packet to a remote host (for example, a response to the packet it has received), it sends as it does normally. The mobile host prepares a packet with its home address as the source, and the address of the remote host as the destination. Although the packet comes from the foreign network, it has the home address of the mobile host. Path 4 of Figure 6.58 shows this step.

Transparency

In this data transfer process, the remote host is unaware of any movement by the mobile host. The remote host sends packets using the home address of the mobile host as the destination address; it receives packets that have the home address of the mobile host as the source address. The movement is totally transparent. The rest of the Internet is not aware of the mobility of the moving host.

The movement of the mobile host is transparent to the rest of the Internet.

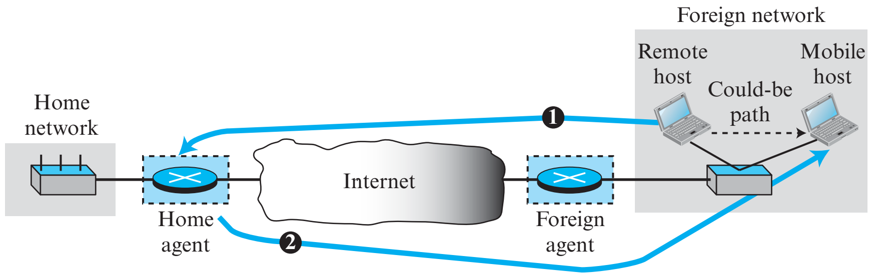
6.3.4 Inefficiency in Mobile IP

Communication involving mobile IP can be inefficient. The inefficiency can be severe or moderate. The severe case is called *double crossing* or *2X*. The moderate case is called *triangle routing* or *dog-leg routing*.

Double Crossing

Double crossing occurs when a remote host communicates with a mobile host that has moved to the same network (or site) as the remote host (see Figure 6.59).

Figure 6.59 *Double crossing*



When the mobile host sends a packet to the remote host, there is no inefficiency; the communication is local. However, when the remote host sends a packet to the mobile host, the packet crosses the Internet twice.

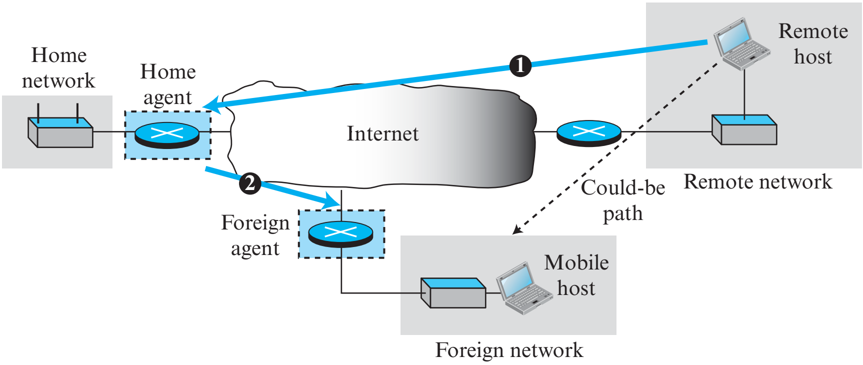
Since a computer usually communicates with other local computers (principle of locality), the inefficiency from double crossing is significant.

Triangle Routing

Triangle routing, the less severe case, occurs when the remote host communicates with a mobile host that is not attached to the same network (or site) as the mobile host.

When the mobile host sends a packet to the remote host, there is no inefficiency. However, when the remote host sends a packet to the mobile host, the packet goes from the remote host to the home agent and then to the mobile host. The packet travels the two sides of a triangle, instead of just one side (see Figure 6.60).

Figure 6.60 *Triangle routing*



Solution

One solution to inefficiency is for the remote host to bind the care-of address to the home address of a mobile host. For example, when a home agent receives the first packet for a mobile host, it forwards the packet to the foreign agent; it could also send an *update binding packet* to the remote host so that future packets to this host could be sent to the care-of address. The remote host can keep this information in a cache.

The problem with this strategy is that the cache entry becomes outdated once the mobile host moves. In this case the home agent needs to send a *warning packet* to the remote host to inform it of the change.

6.4 END-CHAPTER MATERIALS

6.4.1 Further Reading

For more details about subjects discussed in this chapter, we recommend the following books and RFCs. The items in brackets […] refer to the reference list at the end of the text.

Books

Several books cover materials discussed in this chapter, including [Sch 03], [Gas 02]. [For 03], [Sta 04], [Sta 02], [Kei 02], [Jam 03], [AZ 03], [Tan 03], [Cou 01], [Com 06], [G & W 04], and [PD 03].

RFCs

Several RFCs in particular discuss mobile IP: RFC 1701, RFC 2003, RFC 2004, RFC 3024, RFC 3344, and RFC 3775.

6.4.2 Key Terms

access point (AP)
adaptive antenna system (AAS)
ad hoc network
Advanced Mobile Phone System (AMPS)
asynchronous connectionless link (ACL)
basic service set (BSS)
beacon frame
Bluetooth
care-of address
cellular telephony
channelization
code division multiple access (CDMA)
collocated care-of address
complementary code keying (CCK)
customer premises equipment (CPE)
digital AMPS (D-AMPS)
direct sequence spread spectrum (DSSS)
distributed coordination function (DCF)
distributed interframe space (DIFS)
double crossing
extended service set (ESS)
footprint
foreign agent
foreign network
frequency-division multiple access (FDMA)
frequency-hopping spread spectrum (FHSS)
geostationary Earth orbit (GEO)
Global Positioning System (GPS)
Global System for Mobile Communication (GSM)
Globalstar
handoff
high-rate direct-sequence spread spectrum (HR-DSSS)
home address
home agent
home network
interframe space (IFS)
Interim Standard 95 (IS-95)
Interleaved FDMA (IFDMA)
Internet Mobile Communication 2000 (IMT-2000)
Iridium
Logical Link Control and Adaptation Protocol (L2CAP)
low-Earth-orbit (LEO)
medium-Earth-orbit (MEO)
mobile host
mobile switching center (MSC)
multi-carrier CDMA (MC-CDMA)
multiple-input multiple-output (MIMO) antenna
multiuser MIMO (MU-MIMO) antenna
network allocation vector (NAV)
no transition mobility
orthogonal FDMA (OFDMA)
orthogonal frequency-division multiplexing (OFDM)
personal communications system (PCS)
triangulation
piconet
point coordination function (PCF)
pulse position modulation (PPM)
reuse factor
roaming
scalable OFDMA (SOFDMA)
scatternet
short interframe space (SIFS)
Software Defined Radio (SDR)
stationary host
synchronous connection-oriented (SCO) link
TDD-TDMA (time-division duplex TDMA)
Teledesic
time division multiple access (TDMA)
triangle routing
trilateration
Universal Mobile Telecommunication System (UMTS)
Walsh table
Worldwide Interoperability for Microwave Access (WiMAX)

6.4.3 Summary

Wireless LANs became formalized with the IEEE 802.11 standard, which defines two services: basic service set (BSS) and extended service set (ESS). The access method used in the distributed coordination function (DCF) MAC sublayer is CSMA/CA. The access method used in the point coordination function (PCF) MAC sublayer is polling. Bluetooth is a wireless LAN technology that connects devices (called gadgets) in a small area. A Bluetooth network is called a piconet. WiMAX is a wireless access network that may replace DSL and cable in the future.

Cellular telephony provides communication between two devices. One or both may be mobile. A cellular service area is divided into cells. Advanced Mobile Phone System (AMPS) is a first-generation cellular phone system. Digital AMPS (D-AMPS) is a second-generation cellular phone system that is a digital version of AMPS. Global System for Mobile Communication (GSM) is a second-generation cellular phone system used in Europe. Interim Standard 95 (IS-95) is a second-generation cellular phone system based on CDMA and DSSS. The third-generation cellular phone system provides universal personal communication. The fourth generation is the new generation of cellular phones that are becoming popular.

A satellite network uses satellites to provide communication between any points on Earth. A geostationary Earth orbit (GEO) is at the equatorial plane and revolves in phase with Earth's rotation. Global Positioning System (GPS) satellites are medium-Earth-orbit (MEO) satellites that provide time and location information for vehicles and ships. Iridium satellites are low-Earth-orbit (LEO) satellites that provide direct universal voice and data communications for handheld terminals. Teledesic satellites are low-Earth-orbit satellites that will provide universal broadband Internet access.

Mobile IP, designed for mobile communication, is an enhanced version of the Internetworking Protocol (IP). A mobile host has a home address on its home network and a care-of address on its foreign network. When the mobile host is on a foreign network, a home agent relays messages (for the mobile host) to a foreign agent. A foreign agent sends relayed messages to a mobile host.

6.5 PRACTICE SET

6.5.1 Quizzes

A set of interactive quizzes for this chapter can be found on the book website. It is strongly recommended that students take the quizzes to check his/her understanding of the materials before continuing with the practice set.

6.5.2 Questions

Q6-1. What is multipath propagation? What is its effect on wireless networks?

Q6-2. What are some of the reasons that CSMA/CD cannot be used in a wireless LAN?

Q6-3. Compare the medium of a wired LAN with that of a wireless LAN in today's communication environment.

Q6-4. Explain why the MAC protocol is more important in wireless LANs than in wired LANs.

Q6-5. Explain why there is more attenuation in a wireless LAN than in a wired LAN, ignoring the noise and the interference.

Q6-6. Why is SNR in a wireless LAN normally lower than SNR in a wired LAN?

Q6-7. There is no acknowledgment mechanism in CSMA/CD, but we need this mechanism in CSMA/CA. Explain the reason.

Q6-8. What is the purpose of NAV in CSMA/CA?

Q6-9. List some strategies in CSMA/CA that are used to avoid collision.

Q6-10. In a wireless LAN, station A is assigned IFS = 5 milliseconds and station B is assigned IFS = 7 milliseconds. Which station has a higher priority? Explain.

Q6-11. Do the MAC addresses used in an 802.3 (Wired Ethernet) and the MAC addresses used in 802.11 (Wireless Ethernet) belong to two different address spaces?

Q6-12. An AP may connect a wireless network to a wired network. Does the AP need to have two MAC addresses in this case?

Q6-13. An AP in a wireless network plays the same role as a link-layer switch in a wired network. However, a link-layer switch has no MAC address, but an AP normally needs a MAC address. Explain the reason.

Q6-14. What is the reason that Bluetooth is normally called a wireless personal area network (WPAN) instead of a wireless local area network (WLAN)?

Q6-15. Explain why fragmentation is recommended in a wireless LAN.

Q6-16. Explain why we have only one frame type in a wired LAN, but four frame types in a wireless LAN.

Q6-17. What is the actual bandwidth used for communication in a Bluetooth network?

Q6-18. What is the role of the *radio* layer in Bluetooth?

Q6-19. Compare a piconet and a scatternet in the Bluetooth architecture.

Q6-20. Can a piconet have more than eight stations? Explain.

Q6-21. What is the modulation technique in the radio layer of Bluetooth? In other words, how are digital data (bits) changed to analog signals (radio waves)?

Q6-22. What MAC protocol is used in the baseband layer of Bluetooth?

Q6-23. Fill in the blanks. The 83.5 MHz bandwidth in Bluetooth is divided into_____ channels, each of ______ MHz.

Q6-24. What is the spread spectrum technique used by Bluetooth?

Q6-25. Assume that two cellular phone subscribers use FDMA for communication. If each station is allocated a different frequency band, how can two stations communicate with each other?

Q6-26. In TDMA, if each station is allocated a different time slot, how can two stations communicate with each other?

Q6-27. In CDMA, if each station is allocated a different code, how can two stations communicate with each other?

Q6-28. Assume that there are eight stations in a system using CDMA. What is the value of the following inner products of codes?

a. $c_1 . c_1$ **b.** $c_1 . c_4$ **c.** $c_4 . c_1$

Q6-29. What is the role of the *L2CAP* layer in Bluetooth?

Q6-30. In which channelization method do the stations share the available bandwidth of a link in time?

a. FDMA b. TDMA c. CDMA

Q6-31. In which channelization method do the stations share the available bandwidth of a link in frequency?

a. FDMA b. TDMA c. CDMA

Q6-32. In which channelization method are the stations assigned different codes?

a. FDMA b. TDMA c. CDMA

Q6-33. To which generation does each of the following cellular telephony systems belong?

a. AMPS b. D-AMPS c. IS-95

Q6-34. In Mobile IP, is registration required if the mobile host acts as a foreign agent? Explain your answer.

Q6-35. Discuss how the ICMP router solicitation message can be used for agent solicitation.

Q6-36. Explain why the registration request and reply are not directly encapsulated in an IP datagram. Why is there a need for the UDP user datagram?

Q6-37. What is the number of sequences in CDMA for each of the following systems with the given number of stations?

a. 8 stations b. 12 stations c. 28 stations

Q6-38. How many possible codes of size 4 can be created? How many of these codes are orthogonal?

6.5.3 Problems

P6-1. In an 802.11, give the value of the address 3 field in each of the following situations (left bit defines *To DS* and right bit defines *From DS*).

a. 00 b. 01 c. 10 d. 11

P6-2. In an 802.11, give the value of the address 4 field in each of the following situations (left bit defines *To DS* and right bit defines *From DS*).

a. 00 b. 01 c. 10 d. 11

P6-3. In an 802.11, give the value of the address 1 field in each of the following situations (left bit defines *To DS* and right bit defines *From DS*).

a. 00 b. 01 c. 10 d. 11

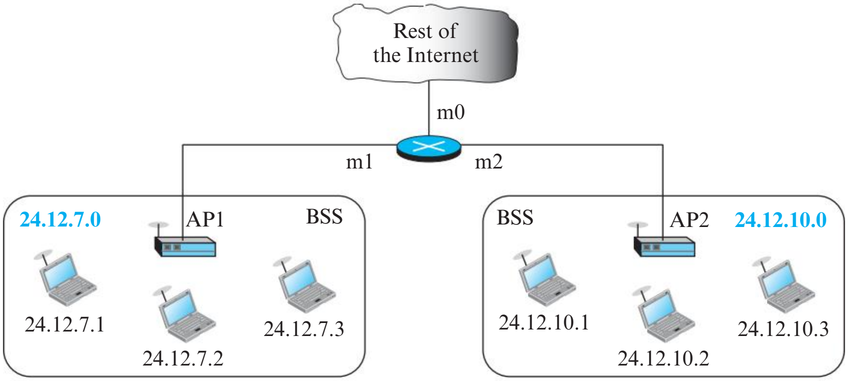
P6-4. In an 802.11, give the value of the address 2 field in each of the following situations (left bit defines *To DS* and right bit defines *From DS*).

a. 00 b. 01 c. 10 d. 11

P6-5. Assume that a frame moves from a wireless network using the 802.11 protocol to a wired network using the 802.3 protocol. Show how the field values in the 802.3 frame are filled with the values of the 802.11 frame. Assume that the transformation occurs at the AP that is in the boundary between the two networks.

P6-6. Assume that two 802.11 wireless networks are connected to the rest of the Internet via a router as shown in Figure 6.61. The router has received an IP datagram with the destination IP address 24.12.7.1 and needs to send it to the corresponding wireless host. Explain the process and describe how the values of address 1, address 2, address 3, and address 4 (see Figure 6.61) are determined in this case.

Figure 6.61 *Problem 6-6*



P6-7. In Figure 6.61 (previous problem), assume that the host with IP address 24.12.10.3 needs to send an IP datagram to the host with IP address 128.41.23.12 somewhere in the world (not shown in the figure). Explain the process and show how the values of address 1, address 2, address 3, and address 4 (see Figure 6.61) are determined in this case.

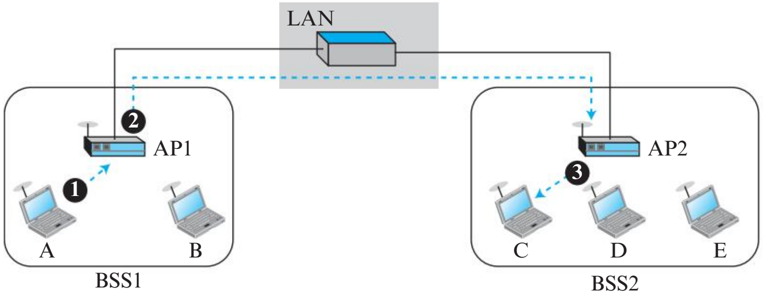
P6-8. A BSS ID (BSSID) is a 48-bit address assigned to a BSS in an 802.11 network. Do some research and find what the use of the BSSID is and how BSSIDs are assigned in ad hoc and infrastructure networks.

P6-9. In a BSS with no AP (ad hoc network), we have five stations: A, B, C, D, and E. Station A needs to send a message to station B. Answer the following questions for the situation where the network is using the DCF protocol:

- **a.** What are the values of the *To DS* and *From DS* bits in the frames exchanged?
- **b.** Which station sends the RTS frame and what is (are) the value(s) of the address field(s) in this frame?
- **c.** Which station sends the CTS frame and what is (are) the value(s) of the address field(s) in this frame?
- **d.** Which station sends the data frame and what is (are) the value(s) of the address field(s) in this frame?
- **e.** Which station sends the ACK frame and what is (are) the value(s) of the address field(s) in this frame?

P6-10. In Figure 6.62, two wireless networks, BSS1 and BSS2, are connected through a wired distribution system (DS), an Ethernet LAN. Assume that station A in BSS1 needs to send a data frame to station C in BSS2. Show the value of addresses in 802.11 and 802.3 frames for three transmissions: from station A to AP1, from AP1 to AP2, and from AP2 to station C. Note that the communication between AP1 and AP2 occurs in a wired environment.

Figure 6.62 *Problem P6-10*



P6-11. Repeat the previous problem (Figure 6.62), but assume that the distribution system is also wireless. AP1 is connected to AP2 through a wireless channel. Show the value of addresses in all communication sections: from station A to AP1, from AP1 to AP2, and from AP2 to station C.

P6-12. Assume that a frame moves from a wireless network using the 802.3 protocol to a wired network using the 802.11 protocol. Show how the field values in the 802.11 frame are filled with the values of the 802.3 frame. Assume that the transformation occurs at the AP that is in the boundary between the two networks.

P6-13. In an 802.11 network, station A sends one data frame (not fragmented) to station B. What would be the value of the D field (in microseconds) that needs to be set for the NAV period in each of the following frames: RTS, CTS, data, and ACK? Assume that the transmission time for RTS, CTS, and ACK is 4 μs each. The transmission time for the data frame is 40 μs and the SIFS duration is set to 1 μs. Ignore the propagation time. Note that each frame needs to set the duration of NAV for the rest of the time the medium needs to be reserved to complete the transaction.

P6-14. In an 802.11 network, station A sends two data fragments to station B. What would be the value of the D field (in microseconds) that needs to be set for the NAV period in each of the following frames: RTS, CTS, data, and ACK? Assume that the transmission time for RTS, CTS, and ACK is 4 μs each. The transmission time for each fragment is 20 μs and the SIFS duration is set to 1 μs. Ignore the propagation time. Note that each frame needs to set the duration of NAV for the rest of the time the medium needs to be reserved to complete the transaction.

P6-15. In an 802.11 network, three stations (A, B, and C) are contending to access the medium. The contention window for each station has 31 slots. Station A randomly picks up the first slot; station B picks up the fifth slot; and station C picks up the twenty-first slot. Show the procedure each station should follow.

P6-16. In an 802.11 network, there are three stations, A, B, and C. Station C is hidden from A, but can be seen (electronically) by B. Now assume that station A needs to send data to station B. Since C is hidden from A, the RTS frame cannot reach C. Explain how station C can find out that the channel is locked by A and that it should refrain from transmitting.

P6-17. Do some research and find out how flow and error control are accomplished in an 802.11 network using the DCF MAC sublayer.

P6-18. In an 802.11 communication, the size of the payload (frame body) is 1200 bytes. The station decides to fragment the frame into three fragments, each of 400 payload bytes. Answer the following questions:

a. What would be the size of the data frame if no fragmentations were done?

b. What is the size of each frame after fragmentation?

c. How many total bytes are sent after fragmentation (ignoring the extra control frames)?

d. How many extra bytes are sent because of fragmentation (again ignoring extra control frames)?

P6-19. Both the IP protocol and the 802.11 project fragment their packets. IP fragments a datagram at the network layer; 802.11 fragments a frame at the data-link layer. Compare and contrast the two fragmentation schemes, using the different fields and subfields used in each protocol.

P6-20. In an 802.11 network, assume that station A has four fragments to send to station B. If the sequence number of the first fragment is selected as 3273, what are the values of the more fragment flag, fragment number, and sequence number?

P6-21. Check to see if the following set of chips can belong to an orthogonal system.

$$[+1, +1, +1, +1] \quad , \quad [+1, -1, -1, +1] \quad , \quad [-1, +1, +1, -1] \quad , \quad [+1, -1, -1, +1]$$

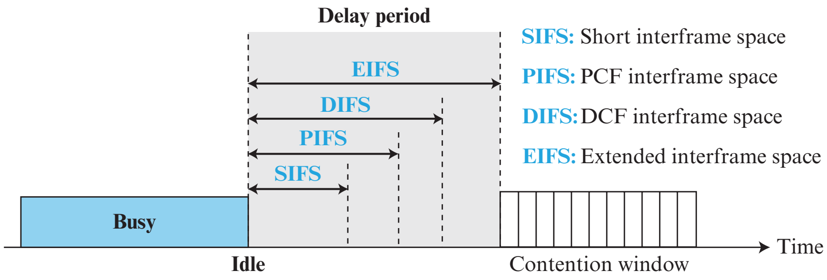
P6-22. Alice and Bob are experimenting with CDMA using a W_2 Walsh table (see Figure 6.34). Alice uses the code [+1, +1] and Bob uses the code [+1, −1]. Assume that they simultaneously send a hexadecimal digit to each other. Alice sends $(6)_{16}$ and Bob sends $(B)_{16}$. Show how they can detect what the other person has sent.

P6-23. Draw a cell pattern with a frequency-reuse factor of 5.

P6-24. Find the efficiency of the AMPS protocol in terms of simultaneous calls per megahertz of bandwidth. In other words, find the number of calls that can be made in 1 MHz bandwidth allocation.

P6-25. An 802.11 network may use four different interframe spaces (IFSs) to delay the transmission of a frame in different situations. This allows low-priority traffic to wait for high-priority traffic when the channel becomes idle. Normally, four different IFSs are used in different implementations, as shown in Figure 6.63. Explain the purpose of these IFSs (you may need to do some research using the Internet).

Figure 6.63 *Problem 6-25*



P6-26. Although an RTS frame defines the value of time that NAV can be effective for the rest of the session, why does the 802.11 project define that other frames used in the session should redefine the rest of the period for NAV?

P6-27. Figure 6.24 shows the frame format of the baseband layer in Bluetooth (802.15). Based on this format, answer the following questions:

a. What is the range of the address domain in a Bluetooth network?

b. How many stations can be active at the same time in a piconet based on the information in the above figure?

P6-28. Check to see if the following set of chips can belong to an orthogonal system.

$$[+1, +1] \quad \text{and} \quad [+1, -1]$$

P6-29. Find the efficiency of the IS-95 protocol in terms of simultaneous calls per megahertz of bandwidth. In other words, find the number of calls that can be made in 1 MHz bandwidth allocation.

P6-30. Use Kepler's law to check the accuracy of a given period and altitude for the GPS satellites.

P6-31. Use Kepler's law to check the accuracy of a given period and altitude for the Globalstar satellites.

P6-32. Redraw Figure 6.58 for the case where the mobile host acts as a foreign agent.

P6-33. Create a home agent advertisement message using 1456 as the sequence number and a lifetime of three hours. Select your own values for the bits in the code field. Calculate and insert the value for the length field.

P6-34. Create a foreign agent advertisement message using 1672 as the sequence number and a lifetime of four hours. Select your own values for the bits in the code field. Use at least three care-of addresses of your choice. Calculate and insert the value for the length field.

P6-35. Find the efficiency of the Q-AMPS protocol in terms of simultaneous calls per megahertz of bandwidth. In other words, find the number of calls that can be made in 1 MHz bandwidth allocation.

P6-36. Find the efficiency of the GSM protocol in terms of simultaneous calls per megahertz of bandwidth. In other words, find the number of calls that can be made in 1 MHz bandwidth allocation.

P6-37. We have the information shown below. Show the contents of the IP datagram header sent from the remote host to the home agent.

Mobile host home address: 130.45.6.7/16
Mobile host care-of address: 14.56.8.9/8
Remote host address: 200.4.7.14/24
Home agent address: 130.45.10.20/16
Foreign agent address: 14.67.34.6/8

6.6 SIMULATION EXPERIMENTS

6.6.1 Applets

We have created some Java applets to show some of the main concepts discussed in this chapter. It is strongly recommended that students activate these applets on the book website and carefully examine the protocols in action.

6.6.2 Lab Assignments

Use the book website to find how you can use a simulator for wireless LANs.

Lab6-1. In this lab, we capture and study wireless frames that are exchanged between a wireless host and the access point. See the book website for a detailed description of this lab.

6.7 PROGRAMMING ASSIGNMENT

Write the source code, compile, and test the following program in one of the programming languages of your choice:

Prg6-1. Write a program to simulate the flow diagram of CSMA/CA in Figure 6.7.